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R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
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Platform: x86_64-w64-mingw32/x64

R 是自由軟體，不提供任何擔保。
在某些條件下歡迎您將其散佈。
用 'license()' 或 'licence()' 來獲得散佈的詳細條件。

R 是個協作計劃，有許多人為之做出了貢獻。
用 'contributors()' 來看詳細的情況以及
用 'citation()' 會告訴您如何在出版品中正確地參照 R 或 R 套件。

用 'demo()' 來看一些示範程式，用 'help()' 來檢視線上輔助檔案，或
用 'help.start()' 透過 HTML 瀏覽器來看輔助檔案。
用 'q()' 離開 R。

[原來儲存的工作空間已還原]

> # CHAPTER 5 Decision Making for Two Samples
> # Refer to CHAPTER 4.
> # § 5-1 Introduction
> # Refer to § 4-1.
> # Figure 5-1 Two independent populations.
> # § 5-2 Inference on the Means of Two Populations, Variances Known
> # Refer to § 4-4.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
載入需要的套件：pacman
> pacman::p_load(HH, BSDA, ggplot2, purrr, dplyr)
> # --- 2. Functions ---
> Pwr.z.test <- function(n, delta, sigma, alpha, alt = "two.sided") {
+ m <- delta / (sigma / sqrt(n))
+ if (alt == "two.sided") {
+ return(1 - pnorm(qnorm(1 - alpha/2) - m) + pnorm(qnorm(alpha/2) - m))
+ } else if (alt == "greater") {
+ return(1 - pnorm(qnorm(1 - alpha) - m))
+ } else { # alt == "less"
+ return(pnorm(qnorm(alpha) - m))
+ }
+ }
> SplSz.z.test <- function(delta, sigma, alpha, pwr, side = 2) {
+ ceiling(((qnorm(1 - alpha/side) + qnorm(pwr)) * sigma / delta)^2)
+ }
> SplSz.z.CI <- function(E, sigma, conf.level, side = 2) {
+ ceiling((qnorm(1 - (1 - conf.level)/side) * sigma / E)^2)
+ }
> # § 5-2.1 Hypothesis Testing on the Difference in Means, Variances Known
> # Refer to § 4-4.1.
> # EXAMPLE 5-1 Paint Drying Time
> # Refer to EXAMPLE 4-3.
> # --- 3. Parameters ---
> (params <- list(Delta0 = 0, sigma = c(8, 8), alpha = 0.05, n = c(10, 10),
+ delta = c(-10, 10), pwr.target = 0.90, E = 6,
+ conf.target = 0.95)) # δ ≡ Δ − Δ₀ with Δ ≡ μ₁ − μ₂
$Delta0
[1] 0

$sigma
[1] 8 8

$alpha
[1] 0.05

$n
[1] 10 10

$delta
[1] -10  10

$pwr.target
[1] 0.9

$E
```

```
[1] 6

$conf.target
[1] 0.95

> delta_range <- seq(-20, 20, length.out = 100) # δ ≡ Δ − Δ₀
> (n_test_sizes <- c(10, 25, 50, 100)) # n_1 = n_2 = n = 10, 25, 50 and 100
[1]  10  25  50 100
> # --- 4. Basic Stats & Data Simulation ---
> (xbar <- c(121, 112))
[1] 121 112
> (se <- params$sigma / sqrt(params$n))
[1] 2.529822 2.529822
> (se0 <- sqrt(sum(params$sigma^2 / params$n)))
[1] 3.577709
> (sd0 <- sqrt(sum(params$sigma^2)))
[1] 11.31371
> (n0 <- (sd0 / se0)^2) # se0 ≡ sd0 / sqrt(n0)
[1] 10
> (z0 <- (-diff(xbar) - params$Delta0) / se0)
[1] 2.515576
> (Delta <- params$Delta0 + params$delta)
[1] -10  10
> set.seed(123)
> Time1_raw <- rnorm(params$n[1], xbar[1], params$sigma[1])
> Time2_raw <- rnorm(params$n[2], xbar[2], params$sigma[2])
> # Rescale to have sample mean xbar
> Time1 <- xbar[1] + (Time1_raw - mean(Time1_raw))
> Time2 <- xbar[2] + (Time2_raw - mean(Time2_raw))
> # --- 5. Two-Sided Test Analysis ---
> # ==========================================================================
> # CASE: TWO-SIDED TEST
> # ==========================================================================
> cat("\n=== TWO-SIDED TEST ANALYSIS ===\n")

=== TWO-SIDED TEST ANALYSIS ===
> (CritVal_2s <- qnorm(c(params$alpha/2, 1 - params$alpha/2)))
[1] -1.959964  1.959964
> (PVal_2s <- 2 * (1 - pnorm(abs(z0))))
[1] 0.01188379
> (CI_2s <- -diff(xbar) + CritVal_2s * se0)
[1]  1.98782 16.01218
> z.test(Time1, Time2, sigma.x = params$sigma[1], sigma.y = params$sigma[2])

        Two-sample z-Test

data:  Time1 and Time2
z = 2.5156, p-value = 0.01188
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
  1.98782 16.01218
sample estimates:
mean of x mean of y
      121       112

> cat(sprintf("Test Statistic (z0): %.2f\nP-Value: %.4f\n", z0, PVal_2s))
Test Statistic (z0): 2.52
P-Value: 0.0119
> cat("Confidence Interval:", "[", CI_2s[1], ",", CI_2s[2], "]\n")
Confidence Interval: [ 1.98782 , 16.01218 ]
> # 5.1 Power at δ = −10, 10:
> Pwr.z.test(n0, params$delta, sd0, params$alpha)
[1] 0.7981762 0.7981762
> # 5.2 Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> power_2s_df <- expand.grid(delta = delta_range, n = n_test_sizes) %>%
+ mutate(power = mapply(Pwr.z.test, n = n, delta = delta,
+ MoreArgs = list(sigma = sd0,
+ alpha = params$alpha)))
> # 5.3 Sample Size Requirements (n_1 = n_2 = n)
> # a. For Hypothesis Testing
> (n_req_2s <- SplSz.z.test(params$delta, sd0, params$alpha,
+ params$pwr.target))
[1] 14 14
> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at Δ = %s: %s\n",
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+ params$pwr.target, paste(Delta, collapse = ", "),
+ paste(n_req_2s, collapse = ", ")))
Required n1 = n2 = n for Power ≥ 0.90 at Δ = -10, 10: 14, 14
> cat("\nPower at Δ =", Delta[1], "when n1 = n2 = n =", n_req_2s[1] - 1,
+ ",", n_req_2s[1], ":",
+ Pwr.z.test(c(n_req_2s[1] - 1, n_req_2s[1]), params$delta[1], sd0,
+ params$alpha),
+ "\n", "\nPower at Δ = ", Delta[2], "when n1 = n2 = n =",
+ n_req_2s[2] - 1, ",", n_req_2s[2], ":",
+ Pwr.z.test(c(n_req_2s[2] - 1, n_req_2s[2]), params$delta[2], sd0,
+ params$alpha),
+ "\n")

Power at Δ = -10 when n1 = n2 = n = 13 , 14 : 0.8900744 0.9110462

Power at Δ =  10 when n1 = n2 = n = 13 , 14 : 0.8900744 0.9110462
> # b. For Confidence Interval
> (SplSz_CI_2s <- SplSz.z.CI(params$E, sd0, params$conf.target))
[1] 14
> cat(sprintf("Required n1 = n2 = n for Error Margin = %s and Confidence ≥
+ %.2f: %d\n",
+ params$E, params$conf.target, SplSz_CI_2s))
Required n1 = n2 = n for Error Margin = 6 and Confidence ≥
0.95: 14
> cat("\nNote: {|X1¯ - X2¯ - (μ1 - μ2)| ≤ E} =
+ {μ1 - μ2 ε [X1¯ - X2¯ - E, X1¯ - X2¯ + E]} and
+ CI = [x1¯ - x2¯ - E, x1¯ - x2¯ + E] = [", -diff(xbar) - params$E,
+ ",",
+ -diff(xbar) + params$E, "].
+ n1 = n2 = n =", SplSz_CI_2s - 1, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_2s[1] * sd0 / sqrt(SplSz_CI_2s - 1), ",",
+ -diff(xbar) + CritVal_2s[2] * sd0 / sqrt(SplSz_CI_2s - 1), "]
+ n1 = n2 = n =", SplSz_CI_2s, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_2s[1] * sd0 / sqrt(SplSz_CI_2s), ",",
+ -diff(xbar) + CritVal_2s[2] * sd0 / sqrt(SplSz_CI_2s), "]\n")

Note: {|X1¯ - X2¯ - (μ1 - μ2)| ≤ E} =
{μ1 - μ2 ε [X1¯ - X2¯ - E, X1¯ - X2¯ + E]} and
CI = [x1¯ - x2¯ - E, x1¯ - x2¯ + E] = [ 3 , 15 ].
n1 = n2 = n = 13 : 95 % CI = [ 2.849911 , 15.15009 ]
n1 = n2 = n = 14 : 95 % CI = [ 3.073626 , 14.92637 ]
> # 5.4 Visualizations
> # a. Hypothesis Testing (Current n1 = n2 = n = 10)
> # Showing the Effect of δ (≡ Δ - Δ₀ with Δ ≡ μ₁ - μ₂) on Power
> normal.and.t.dist(params$Delta0, Delta[1], -diff(xbar), sd0, n0, NA,
+ params$alpha/2, params$alpha/2, T, T, T)
> normal.and.t.dist(params$Delta0, Delta[2], -diff(xbar), sd0, n0, NA,
+ params$alpha/2, params$alpha/2, T, T, T)
> # b. Confidence Interval (Current n1 = n2 = n = 10)
> normal.and.t.dist(NA, NA, -diff(xbar), sd0, n0, NA, params$alpha/2,
+ params$alpha/2, F, T, T, T, "c")
> # c. Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> ggplot(power_2s_df, aes(x = delta, y = power, color = as.factor(n))) +
+ geom_line(linewidth = 1) +
+ geom_hline(yintercept = params$pwr.target, linetype = "dashed",
+ color = "red") +
+ labs(title = "Power Curves for Two-Sample Z-Test (Known Variances)",
+ subtitle =
+ expression(H[0]*": "*Delta* = "*Delta[0]* vs. "*H[1]*": "*Delta* ≠
+ "*Delta[0]),
+ x = expression(delta == Delta - Delta[0]), y = "Power",
+ color = "Sample Size (n)")
> # --- 6. One-Sided (Greater) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (GREATER) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (GREATER) TEST ANALYSIS ===\n")

=== ONE-SIDED (GREATER) TEST ANALYSIS ===
> (CritVal_gr <- qnorm(c(params$alpha, 1)))
[1] -1.644854       Inf
> (PVal_gr <- 1 - pnorm(z0))
[1] 0.005941895
> (CI_gr <- -diff(xbar) + CritVal_gr * se0)
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[1] 3.115193      Inf
> z.test(Time1, Time2, alternative = "greater", sigma.x = params$sigma[1],
+ sigma.y = params$sigma[2])

        Two-sample z-Test

data:  Time1 and Time2
z = 2.5156, p-value = 0.005942
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 3.115193       NA
sample estimates:
mean of x mean of y
      121       112

> cat(sprintf("Test Statistic (z0): %.2f\nP-Value: %.4f\n", z0, PVal_gr))
Test Statistic (z0): 2.52
P-Value: 0.0059
> cat("Confidence Interval:", "[", CI_gr[1], ", ∞)\n")
Confidence Interval: [ 3.115193 , ∞)
> # 6.1 Power at δ = 10:
> Pwr.z.test(n0, params$delta[2], sd0, params$alpha, "greater")
[1] 0.8749757
> # 6.2 Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> power_gr_df <- expand.grid(delta = delta_range, n = n_test_sizes) %>%
+ mutate(power = mapply(Pwr.z.test, n = n, delta = delta,
+ MoreArgs = list(sigma = sd0,
+ alpha = params$alpha,
+ alt = "greater")))
> # 6.3 Sample Size Requirements (n1 = n2 = n)
> # a. For Hypothesis Testing
> (n_req_gr <- SplSz.z.test(params$delta[2], sd0, params$alpha,
+ params$pwr.target, 1))
[1] 11
> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at Δ = %s: %s\n",
+ params$pwr.target, Delta[2], n_req_gr))
Required n1 = n2 = n for Power ≥ 0.90 at Δ = 10: 11
> cat("\nPower at Δ = ", Delta[2], "when n1 = n2 = n =", n_req_2s[2] - 1,
+ ",", n_req_2s[2], ":",
+ Pwr.z.test(c(n_req_2s[2] - 1, n_req_2s[2]), params$delta[2], sd0,
+ params$alpha),
+ "\n")

Power at Δ =  10 when n1 = n2 = n = 13 , 14 : 0.8900744 0.9110462
> # b. For Confidence Interval
> (SplSz_CI_gr <- SplSz.z.CI(params$E, sd0, params$conf.target, 1))
[1] 10
> cat(sprintf("Required n1 = n2 = n for Error Margin = %s and Confidence ≥
+ %.2f: %d\n",
+ params$E, params$conf.target, SplSz_CI_gr))
Required n1 = n2 = n for Error Margin = 6 and Confidence ≥
0.95: 10
> cat("\nNote: {X1¯ − X2¯ − (μ1 − μ2) ≤ E} =
+ {μ1 − μ2 ε [X1¯ − X2¯ − E, ∞)} and
+ CI = [x1¯ − x2¯ − E, ∞) = [", -diff(xbar) - params$E, ",", "∞).
+ n1 = n2 = n =", SplSz_CI_gr - 1, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_gr[1] * sd0 / sqrt(SplSz_CI_gr - 1), ", ∞)\
+ n1 = n2 = n =", SplSz_CI_gr, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_gr[1] * sd0 / sqrt(SplSz_CI_gr), ", ∞)\n")

Note: {X1¯ − X2¯ − (μ1 − μ2) ≤ E} =
{μ1 − μ2 ε [X1¯ − X2¯ − E, ∞)} and
CI = [x1¯ − x2¯ − E, ∞) = [ 3 , ∞).
n1 = n2 = n = 9 : 95 % CI = [ 2.796869 , ∞)
n1 = n2 = n = 10 : 95 % CI = [ 3.115193 , ∞)
> # 6.4 Visualizations
> # a. Hypothesis Testing (Current n1 = n2 = n = 10)
> # Showing the Effect of δ (≡ Δ − Δ₀ with Δ ≡ μ₁ − μ₂) on Power
> normal.and.t.dist(params$Delta0, Delta[2], -diff(xbar), sd0, n0, NA,
+ params$alpha, params$alpha, T, T)
> # b. Confidence Interval (Current n1 = n2 = n = 10)
> normal.and.t.dist(NA, NA, -diff(xbar), sd0, n0, NA, params$alpha,
+ params$alpha, F, F, T, F, "c")
> # c. Power Curves for n1 = n2 = n = 10, 25, 50 and 100
```

```
> ggplot(power_gr_df, aes(x = delta, y = power, color = as.factor(n))) +
+ geom_line(linewidth = 1) +
+ geom_hline(yintercept = params$pwr.target, linetype = "dashed",
+ color = "red") +
+ labs(title = "Power Curves for Two-Sample Z-Test (Known Variances)",
+ subtitle =
+ expression(H[0]*": "*Delta*" ≤ "*Delta[0]*" vs. "*H[1]*": "*Delta*" >
+ "*Delta[0]),
+ x = expression(delta == Delta - Delta[0]), y = "Power",
+ color = "Sample Size (n)")
> # --- 7. One-Sided (Less) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (LESS) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (LESS) TEST ANALYSIS ===\n")

=== ONE-SIDED (LESS) TEST ANALYSIS ===
> (CritVal_le <- qnorm(c(0, 1 - params$alpha)))
[1]     -Inf 1.644854
> (PVal_le <- pnorm(z0))
[1] 0.9940581
> (CI_le <- -diff(xbar) + CritVal_le * se0)
[1]     -Inf 14.88481
> z.test(Time1, Time2, alternative = "less", sigma.x = params$sigma[1],
+ sigma.y = params$sigma[2])

        Two-sample z-Test

data:  Time1 and Time2
z = 2.5156, p-value = 0.9941
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
       NA 14.88481
sample estimates:
mean of x mean of y
      121       112

> cat(sprintf("Test Statistic (z0): %.2f\nP-Value: %.4f\n", z0, PVal_le))
Test Statistic (z0): 2.52
P-Value: 0.9941
> cat("Confidence Interval:", "(-∞,", CI_le[2], "]\n")
Confidence Interval: (-∞, 14.88481 ]
> # 7.1 Power at δ = -10:
> Pwr.z.test(n0, params$delta[1], sd0, params$alpha, "less")
[1] 0.8749757
> # 7.2 Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> power_le_df <- expand.grid(delta = delta_range, n = n_test_sizes) %>%
+ mutate(power = mapply(Pwr.z.test, n = n, delta = delta,
+ MoreArgs = list(sigma = sd0,
+ alpha = params$alpha,
+ alt = "less")))
> # 7.3 Sample Size Requirements (n1 = n2 = n)
> # a. For Hypothesis Testing
> (n_req_le <- SplSz.z.test(params$delta[1], sd0, params$alpha,
+ params$pwr.target, 1))
[1] 11
> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at Δ = %s: %s\n",
+ params$pwr.target, Delta[1], n_req_le))
Required n1 = n2 = n for Power ≥ 0.90 at Δ = -10: 11
> cat("\nPower at Δ =", Delta[1], "when n1 = n2 = n =", n_req_le - 1, ",",
+ n_req_le, ":",
+ Pwr.z.test(c(n_req_le - 1, n_req_le), params$delta[1], sd0, params$alpha,
+ "less"),
+ "\n")

Power at Δ = -10 when n1 = n2 = n = 10 , 11 : 0.8749757 0.9008929
> # b. For Confidence Interval
> (SplSz_CI_le <- SplSz.z.CI(params$E, sd0, params$conf.target, 1))
[1] 10
> cat(sprintf("Required n1 = n2 = n for Error Margin = %s and Confidence ≥
+ %.2f: %d\n",
+ params$E, params$conf.target, SplSz_CI_le))
Required n1 = n2 = n for Error Margin = 6 and Confidence ≥
0.95: 10
```

```
> cat("\nNote: {X1¯ - X2¯ - (μ1 - μ2) ≥ -E} =
+ {μ1 - μ2 ε (-∞, X1¯ - X2¯ + E]} and
+ CI = (-∞, x1¯ - x2¯ + E] = (-∞,", -diff(xbar) + params$E, "].
+ n1 = n2 = n =", SplSz_CI_le - 1, ":", 100 * params$conf.target,
+ "% CI = (-∞,",
+ -diff(xbar) + CritVal_le[2] * sd0 / sqrt(SplSz_CI_le - 1), "]
+ n1 = n2 = n =", SplSz_CI_le, ":", 100 * params$conf.target, "% CI = (-∞,",
+ -diff(xbar) + CritVal_le[2] * sd0 / sqrt(SplSz_CI_le), "]\n")

Note: {X1¯ - X2¯ - (μ1 - μ2) ≥ -E} =
{μ1 - μ2 ε (-∞, X1¯ - X2¯ + E]} and
CI = (-∞, x1¯ - x2¯ + E] = (-∞, 15 ].
n1 = n2 = n = 9 : 95 % CI = (-∞, 15.20313 ]
n1 = n2 = n = 10 : 95 % CI = (-∞, 14.88481 ]
> # 7.4 Visualizations
> # a. Hypothesis Testing (Current n1 = n2 = n = 10)
> # Showing the Effect of δ (≡ Δ - Δ₀ with Δ ≡ μ₁ - μ₂) on Power
> normal.and.t.dist(params$Delta0, Delta[1], -diff(xbar), sd0, n0, NA,
+ params$alpha, params$alpha, T, T, T, F)
> # b. Confidence Interval (Current n1 = n2 = n = 10)
> normal.and.t.dist(NA, NA, -diff(xbar), sd0, n0, NA, params$alpha,
+ params$alpha, F, T, F, T, "c")
> # c. Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> ggplot(power_le_df, aes(x = delta, y = power, color = as.factor(n))) +
+ geom_line(linewidth = 1) +
+ geom_hline(yintercept = params$pwr.target, linetype = "dashed",
+ color = "red") +
+ labs(title = "Power Curves for Two-Sample Z-Test (Known Variances)",
+ subtitle =
+ expression(H[0]*": "*Delta*" ≥ "*Delta[0]*" vs. "*H[1]*": "*Delta*" <
+ "*Delta[0]),
+ x = expression(delta == Delta - Delta[0]), y = "Power",
+ color = "Sample Size (n)")
> # § 5-2.2 Type II Error and Choice of Sample Size
> # Refer to § 4-4.2.
> # EXAMPLE 5-2 Paint Drying Time
> # Refer to EXAMPLES 4-4 and 5-1.
> # § 5-2.3 Confidence Interval on the Difference in Means, Variances Known
> # Refer to § 4-4.5.
> # EXAMPLE 5-3 Aircraft Spars
> # Table 5-1 Tensile Strength Test Result for Aluminum Spars
> # Refer to EXAMPLE 5-1.
> # --- 3*. Parameters ---
> (params <- list(Delta0 = 0, sigma = c(1.0, 1.5), alpha = 0.1,
+ n = c(10, 12), delta = c(-1, 1), pwr.target = 0.9, E = 0.6,
+ conf.target = 0.9)) # δ ≡ Δ - Δ₀ with Δ ≡ μ₁ - μ₂
$Delta0
[1] 0

$sigma
[1] 1.0 1.5

$alpha
[1] 0.1

$n
[1] 10 12

$delta
[1] -1  1

$pwr.target
[1] 0.9

$E
[1] 0.6

$conf.target
[1] 0.9

> delta_range <- seq(-3, 3, length.out = 100) # δ ≡ Δ - Δ₀
> (n_test_sizes <- c(10, 25, 50, 100)) # n_1 = n_2 = n = 10, 25, 50 and 100
[1]  10  25  50 100
> # --- 4*. Basic Stats & Data Simulation ---
```

```
> (xbar <- c(87.6, 74.5))
[1] 87.6 74.5
> (se <- params$sigma / sqrt(params$n))
[1] 0.3162278 0.4330127
> (se0 <- sqrt(sum(params$sigma^2 / params$n)))
[1] 0.5361903
> (sd0 <- sqrt(sum(params$sigma^2)))
[1] 1.802776
> (n0 <- (sd0 / se0)^2) # se0 ≡ sd0 / sqrt(n0)
[1] 11.30435
> (z0 <- (-diff(xbar) - params$Delta0) / se0)
[1] 24.43163
> (Delta <- params$Delta0 + params$delta)
[1] -1  1
> set.seed(123)
> TS1_raw <- rnorm(params$n[1], xbar[1], params$sigma[1])
> TS2_raw <- rnorm(params$n[2], xbar[2], params$sigma[2])
> # Rescale to have sample mean xbar
> TS1 <- xbar[1] + (TS1_raw - mean(TS1_raw))
> TS2 <- xbar[2] + (TS2_raw - mean(TS2_raw))
> # --- 5*. Two-Sided Test Analysis ---
> # ==========================================================================
> # CASE: TWO-SIDED TEST
> # ==========================================================================
> cat("\n=== TWO-SIDED TEST ANALYSIS ===\n")

=== TWO-SIDED TEST ANALYSIS ===
> (CritVal_2s <- qnorm(c(params$alpha/2, 1 - params$alpha/2)))
[1] -1.644854  1.644854
> (PVal_2s <- 2 * (1 - pnorm(abs(z0))))
[1] 0
> (CI_2s <- -diff(xbar) + CritVal_2s * se0)
[1] 12.21805 13.98195
> z.test(TS1, TS2, sigma.x = params$sigma[1], sigma.y = params$sigma[2],
+ conf.level = 0.9)

        Two-sample z-Test

data:  TS1 and TS2
z = 24.432, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 12.21805 13.98195
sample estimates:
mean of x mean of y
     87.6      74.5

> cat(sprintf("Test Statistic (z0): %.2f\nP-Value: %.4f\n", z0, PVal_2s))
Test Statistic (z0): 24.43
P-Value: 0.0000
> cat("Confidence Interval:", "[", CI_2s[1], ",", CI_2s[2], "]\n")
Confidence Interval: [ 12.21805 , 13.98195 ]
> # 5*.1 Power at δ = -1, 1:
> Pwr.z.test(n0, params$delta, sd0, params$alpha)
[1] 0.5873493 0.5873493
> # 5*.2 Power Curves for n_1 = n_2 = n = 10, 25, 50 and 100
> power_2s_df <- expand.grid(delta = delta_range, n = n_test_sizes) %>%
+ mutate(power = mapply(Pwr.z.test, n = n, delta = delta,
+ MoreArgs = list(sigma = sd0,
+ alpha = params$alpha)))
> # 5*.3 Sample Size Requirements (n_1 = n_2 = n)
> # a. For Hypothesis Testing
> (n_req_2s <- SplSz.z.test(params$delta, sd0, params$alpha,
+ params$pwr.target))
[1] 28 28
> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at Δ = %s: %s\n",
+ params$pwr.target, paste(Delta, collapse = ", "),
+ paste(n_req_2s, collapse = ", ")))
Required n1 = n2 = n for Power ≥ 0.90 at Δ = -1, 1: 28, 28
> cat("\nPower at Δ =", Delta[1], "when n1 = n2 = n =", n_req_2s[1] - 1,
+ ",", n_req_2s[1], ":",
+ Pwr.z.test(c(n_req_2s[1] - 1, n_req_2s[1]), params$delta[1], sd0,
+ params$alpha),
+ "\n", "\nPower at Δ = ", Delta[2], "when n1 = n2 = n =",
```

```
+ n_req_2s[2] - 1, ",", n_req_2s[2], ":",
+ Pwr.z.test(c(n_req_2s[2] - 1, n_req_2s[2]), params$delta[2], sd0,
+ params$alpha),
+ "\n")

Power at Δ = -1 when n1 = n2 = n = 27 , 28 : 0.8920435 0.9015367

Power at Δ =  1 when n1 = n2 = n = 27 , 28 : 0.8920435 0.9015367
> # b. For Confidence Interval
> (SplSz_CI_2s <- SplSz.z.CI(params$E, sd0, params$conf.target))
[1] 25
> cat(sprintf("Required n1 = n2 = n for Error Margin = %s and Confidence ≥
+ %.2f: %d\n",
+ params$E, params$conf.target, SplSz_CI_2s))
Required n1 = n2 = n for Error Margin = 0.6 and Confidence ≥
0.90: 25
> cat("\nNote: {|X1¯ - X2¯ - (μ1 - μ2)| ≤ E} =
+ {μ1 - μ2 ε [X1¯ - X2¯ - E, X1¯ - X2¯ + E]} and
+ CI = [x1¯ - x2¯ - E, x1¯ - x2¯ + E] = [", -diff(xbar) - params$E,
+ ",",
+ -diff(xbar) + params$E, "].
+ n1 = n2 = n =", SplSz_CI_2s - 1, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_2s[1] * sd0 / sqrt(SplSz_CI_2s - 1), ",",
+ -diff(xbar) + CritVal_2s[2] * sd0 / sqrt(SplSz_CI_2s - 1), "]
+ n1 = n2 = n =", SplSz_CI_2s, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_2s[1] * sd0 / sqrt(SplSz_CI_2s), ",",
+ -diff(xbar) + CritVal_2s[2] * sd0 / sqrt(SplSz_CI_2s), "]\n")

Note: {|X1¯ - X2¯ - (μ1 - μ2)| ≤ E} =
{μ1 - μ2 ε [X1¯ - X2¯ - E, X1¯ - X2¯ + E]} and
CI = [x1¯ - x2¯ - E, x1¯ - x2¯ + E] = [ 12.5 , 13.7 ].
n1 = n2 = n = 24 : 90 % CI = [ 12.49471 , 13.70529 ]
n1 = n2 = n = 25 : 90 % CI = [ 12.50694 , 13.69306 ]
> # 5*.4 Visualizations
> # a. Hypothesis Testing (Current n1 = 10 and n2 = 12)
> # Showing the Effect of δ (≡ Δ - Δ₀ with Δ ≡ μ₁ - μ₂) on Power
> normal.and.t.dist(params$Delta0, Delta[1], -diff(xbar), sd0, n0, NA,
+ params$alpha/2, params$alpha/2, T, T, T)
> normal.and.t.dist(params$Delta0, Delta[2], -diff(xbar), sd0, n0, NA,
+ params$alpha/2, params$alpha/2, T, T, T)
> # b. Confidence Interval (Current n1 = 10 and n2 = 12)
> normal.and.t.dist(NA, NA, -diff(xbar), sd0, n0, NA, params$alpha/2,
+ params$alpha/2, F, T, T, T, "c")
> # c. Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> ggplot(power_2s_df, aes(x = delta, y = power, color = as.factor(n))) +
+ geom_line(linewidth = 1) +
+ geom_hline(yintercept = params$pwr.target, linetype = "dashed",
+ color = "red") +
+ labs(title = "Power Curves for Two-Sample Z-Test (Known Variances)",
+ subtitle =
+ expression(H[0]*": "*Delta* = "*Delta[0]*" vs. "*H[1]*": "*Delta* ≠
+ "*Delta[0]),
+ x = expression(delta == Delta - Delta[0]), y = "Power",
+ color = "Sample Size (n)")
> # --- 6*. One-Sided (Greater) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (GREATER) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (GREATER) TEST ANALYSIS ===\n")

=== ONE-SIDED (GREATER) TEST ANALYSIS ===
> (CritVal_gr <- qnorm(c(params$alpha, 1)))
[1] -1.281552       Inf
> (PVal_gr <- 1 - pnorm(z0))
[1] 0
> (CI_gr <- -diff(xbar) + CritVal_gr * se0)
[1] 12.41284      Inf
> z.test(TS1, TS2, alternative = "greater", sigma.x = params$sigma[1],
+ sigma.y = params$sigma[2], conf.level = 0.9)

        Two-sample z-Test

data:  TS1 and TS2
z = 24.432, p-value < 2.2e-16
```

```
alternative hypothesis: true difference in means is greater than 0
90 percent confidence interval:
 12.41284       NA
sample estimates:
mean of x mean of y
     87.6      74.5

> cat(sprintf("Test Statistic (z0): %.2f\nP-Value: %.4f\n", z0, PVal_gr))
Test Statistic (z0): 24.43
P-Value: 0.0000
> cat("Confidence Interval:", "[", CI_gr[1], ", ∞)\n")
Confidence Interval: [ 12.41284 , ∞)
> # 6*.1 Power at δ = 1:
> Pwr.z.test(n0, params$delta[2], sd0, params$alpha, "greater")
[1] 0.7202075
> # 6*.2 Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> power_gr_df <- expand.grid(delta = delta_range, n = n_test_sizes) %>%
+ mutate(power = mapply(Pwr.z.test, n = n, delta = delta,
+ MoreArgs = list(sigma = sd0,
+ alpha = params$alpha,
+ alt = "greater")))
> # 6*.3 Sample Size Requirements (n1 = n2 = n)
> # a. For Hypothesis Testing
> (n_req_gr <- SplSz.z.test(params$delta[2], sd0, params$alpha,
+ params$pwr.target, 1))
[1] 22
> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at Δ = %s: %s\n",
+ params$pwr.target, Delta[2], n_req_gr))
Required n1 = n2 = n for Power ≥ 0.90 at Δ = 1: 22
> # b. For Confidence Interval
> (SplSz_CI_gr <- SplSz.z.CI(params$E, sd0, params$conf.target, 1))
[1] 15
> cat(sprintf("Required n1 = n2 = n for Error Margin = %s and Confidence ≥
+ %.2f: %d\n",
+ params$E, params$conf.target, SplSz_CI_gr))
Required n1 = n2 = n for Error Margin = 0.6 and Confidence ≥
0.90: 15
> cat("\nNote: {X1¯ − X2¯ − (μ1 − μ2) ≤ E} =
+ {μ1 − μ2 ε [X1¯ − X2¯ − E, ∞)} and
+ CI = [x1¯ − x2¯ − E, ∞) = [", -diff(xbar) - params$E, ",", "∞).
+ n1 = n2 = n =", SplSz_CI_gr - 1, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_gr[1] * sd0 / sqrt(SplSz_CI_gr - 1), ", ∞)\
+ n1 = n2 = n =", SplSz_CI_gr, ":", 100 * params$conf.target, "% CI = [",
+ -diff(xbar) + CritVal_gr[1] * sd0 / sqrt(SplSz_CI_gr), ", ∞)\n")

Note: {X1¯ − X2¯ − (μ1 − μ2) ≤ E} =
{μ1 − μ2 ε [X1¯ − X2¯ − E, ∞)} and
CI = [x1¯ − x2¯ − E, ∞) = [ 12.5 , ∞).
n1 = n2 = n = 14 : 90 % CI = [ 12.48253 , ∞)
n1 = n2 = n = 15 : 90 % CI = [ 12.50347 , ∞)
> # 6*.4 Visualizations
> # a. Hypothesis Testing (Current n1 = 10 and n2 = 12)
> # Showing the Effect of δ (≡ Δ − Δ₀ with Δ ≡ μ₁ − μ₂) on Power
> normal.and.t.dist(params$Delta0, Delta[2], -diff(xbar), sd0, n0, NA,
+ params$alpha, params$alpha, T, T)
> # b. Confidence Interval (Current n1 = 10 and n2 = 12)
> normal.and.t.dist(NA, NA, -diff(xbar), sd0, n0, NA, params$alpha,
+ params$alpha, F, F, T, F, "c")
> # c. Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> ggplot(power_gr_df, aes(x = delta, y = power, color = as.factor(n))) +
+ geom_line(linewidth = 1) +
+ geom_hline(yintercept = params$pwr.target, linetype = "dashed",
+ color = "red") +
+ labs(title = "Power Curves for Two-Sample Z-Test (Known Variances)",
+ subtitle =
+ expression(H[0]*": "*Delta*" ≤ "*Delta[0]*" vs. "*H[1]*": "*Delta*" >
+ "*Delta[0]),
+ x = expression(delta == Delta - Delta[0]), y = "Power",
+ color = "Sample Size (n)")
> # --- 7*. One-Sided (Less) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (LESS) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (LESS) TEST ANALYSIS ===\n")
```

```
=== ONE-SIDED (LESS) TEST ANALYSIS ===
> (CritVal_le <- qnorm(c(0, 1 - params$alpha)))
[1]     -Inf 1.281552
> (PVal_le <- pnorm(z0))
[1] 1
> (CI_le <- -diff(xbar) + CritVal_le * se0)
[1]     -Inf 13.78716
> z.test(TS1, TS2, alternative = "less", sigma.x = params$sigma[1],
+ sigma.y = params$sigma[2], conf.level = 0.9)

        Two-sample z-Test

data:  TS1 and TS2
z = 24.432, p-value = 1
alternative hypothesis: true difference in means is less than 0
90 percent confidence interval:
       NA 13.78716
sample estimates:
mean of x mean of y
     87.6      74.5

> cat(sprintf("Test Statistic (z0): %.2f\nP-Value: %.4f\n", z0, PVal_le))
Test Statistic (z0): 24.43
P-Value: 1.0000
> cat("Confidence Interval:", "(-∞,", CI_le[2], "]\n")
Confidence Interval: (-∞, 13.78716 ]
> # 7*.1 Power at δ = -1:
> Pwr.z.test(n0, params$delta[1], sd0, params$alpha, "less")
[1] 0.7202075
> # 7*.2 Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> power_le_df <- expand.grid(delta = delta_range, n = n_test_sizes) %>%
+ mutate(power = mapply(Pwr.z.test, n = n, delta = delta,
+ MoreArgs = list(sigma = sd0,
+ alpha = params$alpha,
+ alt = "less")))
> # 7*.3 Sample Size Requirements (n1 = n2 = n)
> # a. For Hypothesis Testing
> (n_req_le <- SplSz.z.test(params$delta[1], sd0, params$alpha,
+ params$pwr.target, 1))
[1] 22
> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at Δ = %s: %s\n",
+ params$pwr.target, Delta[1], n_req_le))
Required n1 = n2 = n for Power ≥ 0.90 at Δ = -1: 22
> cat("\nPower at Δ =", Delta[1], "when n1 = n2 = n =", n_req_le - 1, ",",
+ n_req_le, ":",
+ Pwr.z.test(c(n_req_le - 1, n_req_le), params$delta[1], sd0, params$alpha,
+ "less"),
+ "\n")

Power at Δ = -1 when n1 = n2 = n = 21 , 22 : 0.8962382 0.9066197
> # b. For Confidence Interval
> (SplSz_CI_le <- SplSz.z.CI(params$E, sd0, params$conf.target, 1))
[1] 15
> cat(sprintf("Required n1 = n2 = n for Error Margin = %s and Confidence ≥
+ %.2f: %d\n",
+ params$E, params$conf.target, SplSz_CI_le))
Required n1 = n2 = n for Error Margin = 0.6 and Confidence ≥
0.90: 15
> cat("\nNote: {X1¯ - X2¯ - (μ1 - μ2) ≥ -E} =
+ {μ1 - μ2 ε (-∞, X1¯ - X2¯ + E]} and
+ CI = (-∞, x1¯ - x2¯ + E] = (-∞,", -diff(xbar) + params$E, "].
+ n1 = n2 = n =", SplSz_CI_le - 1, ":", 100 * params$conf.target,
+ "% CI = (-∞,",
+ -diff(xbar) + CritVal_le[2] * sd0 / sqrt(SplSz_CI_le - 1), "]
+ n1 = n2 = n =", SplSz_CI_le, ":", 100 * params$conf.target, "% CI = (-∞,",
+ -diff(xbar) + CritVal_le[2] * sd0 / sqrt(SplSz_CI_le), "]\n")

Note: {X1¯ - X2¯ - (μ1 - μ2) ≥ -E} =
{μ1 - μ2 ε (-∞, X1¯ - X2¯ + E]} and
CI = (-∞, x1¯ - x2¯ + E] = (-∞, 13.7 ].
n1 = n2 = n = 14 : 90 % CI = (-∞, 13.71747 ]
n1 = n2 = n = 15 : 90 % CI = (-∞, 13.69653 ]
> # 7*.4 Visualizations
```

```
> # a. Hypothesis Testing (Current n1 = 10 and n2 = 12)
> # Showing the Effect of δ (≡ Δ − Δ₀ with Δ ≡ μ₁ − μ₂) on Power
> normal.and.t.dist(params$Delta0, Delta[1], -diff(xbar), sd0, n0, NA,
+ params$alpha, params$alpha, T, T, T, F)
> # b. Confidence Interval (Current n1 = 10 and n2 = 12)
> normal.and.t.dist(NA, NA, -diff(xbar), sd0, n0, NA, params$alpha,
+ params$alpha, F, T, F, T, "c")
> # c. Power Curves for n1 = n2 = n = 10, 25, 50 and 100
> ggplot(power_le_df, aes(x = delta, y = power, color = as.factor(n))) +
+ geom_line(linewidth = 1) +
+ geom_hline(yintercept = params$pwr.target, linetype = "dashed",
+ color = "red") +
+ labs(title = "Power Curves for Two-Sample Z-Test (Known Variances)",
+ subtitle =
+ expression(H[0]*": "*Delta*" ≥ "*Delta[0]*" vs. "*H[1]*": "*Delta*" <
+ "*Delta[0]),
+ x = expression(delta == Delta - Delta[0]), y = "Power",
+ color = "Sample Size (n)")
> # EXERCISES FOR § 5-2
> # § 5-3 Inference on the Means of Two Populations, Variances Unknown
> # Refer to § 4-5.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(car, HH)
> # --- 2. Function ---
> Pwr.t.test <- function (n, delta, sigma, alpha, alt = "two.sided") {
+ df <- n - 1
+ ncp <- delta / (sigma / sqrt(n))
+ if (alt == "two.sided") {
+ return(1 - pt(qt(1 - alpha/2, df), df, ncp) +
+ pt(qt(alpha/2, df), df, ncp))
+ } else if (alt == "greater") {
+ return(1 - pt(qt(1 - alpha, df), df, ncp))
+ } else { # alt == "less"
+ return(pt(qt(alpha, df), df, ncp))
+ }
+ }
> # § 5-3.1 Hypothesis Testing on the Difference in Means
> # Case 1: σ_1² = σ_2² = σ²
> # EXAMPLE 5-4 Chemical Process Yield
> # Table 5-2 Catalyst Yield Data (Percent) Example 5-4
> # Refer to EXAMPLE 4-7.
> # --- 3. Data & Parameters ---
> (Cat1 <- c(91.50, 94.18, 92.18, 95.39, 91.79, 89.07, 94.72, 89.21))
[1] 91.50 94.18 92.18 95.39 91.79 89.07 94.72 89.21
> (Cat2 <- c(89.19, 90.95, 90.46, 93.21, 97.19, 97.04, 91.07, 92.75))
[1] 89.19 90.95 90.46 93.21 97.19 97.04 91.07 92.75
> (params <- list(Delta0 = 0, alpha = 0.05,
+ n = c(length(Cat1), length(Cat2)), delta = c(-4, 4),
+ pwr.target = 0.85)) # δ ≡ Δ − Δ₀ with Δ ≡ μ₁ − μ₂
$Delta0
[1] 0

$alpha
[1] 0.05

$n
[1] 8 8

$delta
[1] -4  4

$pwr.target
[1] 0.85

> # --- 4. Basic Stats, Normality Test, Box Plots & Variance Test ---
> summary(Cat1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  89.07   90.93   91.98   92.25   94.31   95.39
> summary(Cat2)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  89.19   90.83   91.91   92.73   94.17   97.19
> (xbar <- c(mean(Cat1), mean(Cat2)))
[1] 92.2550 92.7325
```

```
> (s <- c(sd(Cat1), sd(Cat2)))
[1] 2.385019 2.983453
> (df <- params$n - 1)
[1] 7 7
> (df_Case1 <- sum(df))
[1] 14
> (s_p <- sqrt(sum(df * s^2) / df_Case1))
[1] 2.700862
> (se_Case1 <- s_p * sqrt(sum(1 / params$n)))
[1] 1.350431
> (t0 <- (-diff(xbar) - params$Delta0) / se_Case1)
[1] -0.3535909
> (n_Case1 <- df_Case1 + 1)
[1] 15
> (s_Case1 <- sqrt(n_Case1) * se_Case1)
[1] 5.230196
> (Delta <- params$Delta0 + params$delta)
[1] -4  4
> # Normality Test
> shapiro.test(Cat1)

        Shapiro-Wilk normality test

data:  Cat1
W = 0.92171, p-value = 0.4439

> shapiro.test(Cat2)

        Shapiro-Wilk normality test

data:  Cat2
W = 0.88182, p-value = 0.196

> Yield <- c(Cat1, Cat2)
> Type <- c(rep(1, params$n[1]), rep(2, params$n[2]))
> Type <- factor(Type, c("1", "2"), c("Catalyst 1", "Catalyst 2"))
> qqPlot(Yield ~ Type, ylab = "Yield (%)")
> # Comparative Box Plots
> sp(Yield ~ Type, xlab = "Catalyst type", ylab = "Yield (%)")
> # Variance Test
> var.test(Cat1, Cat2)

        F test to compare two variances

data:  Cat1 and Cat2
F = 0.63907, num df = 7, denom df = 7, p-value = 0.5691
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1279433 3.1920724
sample estimates:
ratio of variances
         0.6390651

> # --- 5. Two-Sided Test Analysis ---
> # ==========================================================================
> # CASE: TWO-SIDED TEST
> # ==========================================================================
> cat("\n=== TWO-SIDED TEST ANALYSIS ===\n")

=== TWO-SIDED TEST ANALYSIS ===
> (CritVal_2s <- qt(c(params$alpha/2, 1 - params$alpha/2), df_Case1))
[1] -2.144787  2.144787
> (PVal_2s <- 2 * (1 - pt(abs(t0), df_Case1)))
[1] 0.7289136
> (CI_2s <- -diff(xbar) + CritVal_2s * se_Case1)
[1] -3.373886  2.418886
> t.test(Cat1, Cat2, var.equal = TRUE)

        Two Sample t-test

data:  Cat1 and Cat2
t = -0.35359, df = 14, p-value = 0.7289
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
```

```
 -3.373886  2.418886
sample estimates:
mean of x mean of y
  92.2550   92.7325

> cat(sprintf("Test Statistic (t0): %.3f\nP-Value: %.4f\n", t0, PVal_2s))
Test Statistic (t0): -0.354
P-Value: 0.7289
> cat("Confidence Interval:", "[", CI_2s[1], ",", CI_2s[2], "]\n")
Confidence Interval: [ -3.373886 , 2.418886 ]
> # 5.1 Power at δ = −4, 4:
> Pwr.t.test(n_Case1, params$delta, s_Case1, params$alpha)
[1] 0.7864591 0.7864591
> power.t.test(n = params$n[1], delta = params$delta, sd = s_p,
+ sig.level = params$alpha)

     Two-sample t test power calculation

              n = 8
          delta = 4, 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.786458, 0.786458
    alternative = two.sided

NOTE: n is number in *each* group

> # 5.2 Sample Size Requirements (n_1 = n_2 = n)
> power.t.test(delta = params$delta[1], sd = s_p, sig.level = params$alpha,
+ power = params$pwr.target, type = "two.sample",
+ alternative = "two.sided")

     Two-sample t test power calculation

              n = 9.257032
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = two.sided

NOTE: n is number in *each* group

> power.t.test(delta = params$delta[2], sd = s_p, sig.level = params$alpha,
+ power = params$pwr.target)

     Two-sample t test power calculation

              n = 9.257032
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = two.sided

NOTE: n is number in *each* group

> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at δ = %s: 10, 10\n",
+ params$pwr.target, paste(params$delta, collapse = ", ")))
Required n1 = n2 = n for Power ≥ 0.85 at δ = -4, 4: 10, 10
> power.t.test(n = c(9, 10), delta = params$delta[1], sd = s_p,
+ sig.level = params$alph)

     Two-sample t test power calculation

              n = 9, 10
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.8385638, 0.8791116
    alternative = two.sided

NOTE: n is number in *each* group
```

```
> power.t.test(n = c(9, 10), delta = params$delta[2], sd = s_p,
+ sig.level = params$alph)

     Two-sample t test power calculation

              n = 9, 10
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.8385638, 0.8791116
    alternative = two.sided

NOTE: n is number in *each* group

> # 5.3 Visualizations
> # a. Hypothesis Testing (Current n_1 = n_2 = n = 8)
> # Showing the Effect of δ (≡ Δ − Δ₀) on Power
> NTplot(t.test(Cat1, Cat2, var.equal = TRUE), mean1 = Delta[1], zaxis = T,
+ power = T)
> NTplot(t.test(Cat1, Cat2, var.equal = TRUE), mean1 = Delta[2], zaxis = T,
+ power = T)
> # b. Confidence Interval (Current n_1 = n_2 = n = 8)
> NTplot(t.test(Cat1, Cat2, var.equal = TRUE), type = "confidence",
+ zaxis = T)
> # c. Comparison of Sample Sizes (Targeting Required n)
> # Plotting n − 1 vs. required n_1 = n_2 = n = 10 at δ = −4, 4:
> power.t.test(delta = params$delta[1], sd = s_p, sig.level = params$alph,
+ power = params$pwr.target)

     Two-sample t test power calculation

              n = 9.257032
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = two.sided

NOTE: n is number in *each* group

> NTplot(power.t.test(n = 9, delta = params$delta[1], sd = s_p,
+ sig.level = params$alph), zaxis = T, power = T)
> NTplot(power.t.test(n = 10, delta = params$delta[1], sd = s_p,
+ sig.level = params$alph), zaxis = T, power = T)
> power.t.test(delta = params$delta[2], sd = s_p, sig.level = params$alph,
+ power = params$pwr.target)

     Two-sample t test power calculation

              n = 9.257032
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = two.sided

NOTE: n is number in *each* group

> NTplot(power.t.test(n = 9, delta = params$delta[2], sd = s_p,
+ sig.level = params$alph), zaxis = T, power = T)
> NTplot(power.t.test(n = 10, delta = params$delta[2], sd = s_p,
+ sig.level = params$alph), zaxis = T, power = T)
> # --- 6. One-Sided (Greater) Test Analysis ---
> # ========================================================================
> # CASE: ONE-SIDED (GREATER) TEST
> # ========================================================================
> cat("\n=== ONE-SIDED (GREATER) TEST ANALYSIS ===\n")

=== ONE-SIDED (GREATER) TEST ANALYSIS ===
> (CritVal_gr <- qt(c(params$alpha, 1), df_Case1))
[1] -1.76131      Inf
> (PVal_gr <- 1 - pt(t0, df_Case1))
[1] 0.6355432
> (CI_gr <- -diff(xbar) + CritVal_gr * se_Case1)
```

```
[1] -2.856027       Inf
> t.test(Cat1, Cat2, alternative = "greater", var.equal = TRUE)

        Two Sample t-test

data:  Cat1 and Cat2
t = -0.35359, df = 14, p-value = 0.6355
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -2.856027       Inf
sample estimates:
mean of x mean of y
  92.2550   92.7325

> cat(sprintf("Test Statistic (t0): %.3f\nP-Value: %.4f\n", t0, PVal_gr))
Test Statistic (t0): -0.354
P-Value: 0.6355
> cat("Confidence Interval:", "[", CI_gr[1], ", ∞)\n")
Confidence Interval: [ -2.856027 , ∞)
> # 6.1 Power at δ = 4:
> Pwr.t.test(n_Case1, params$delta[2], s_Case1, params$alpha, "greater")
[1] 0.8789012
> power.t.test(n = params$n[1], delta = params$delta[2], sd = s_p,
+ sig.level = params$alpha, alternative = "one.sided")

     Two-sample t test power calculation

              n = 8
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.8789012
    alternative = one.sided

NOTE: n is number in *each* group

> # 6.2 Sample Size Requirements (n_1 = n_2 = n)
> power.t.test(delta = params$delta[2], sd = s_p, sig.level = params$alpha,
+ power = params$pwr.target, alternative = "one.sided")

     Two-sample t test power calculation

              n = 7.341304
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = one.sided

NOTE: n is number in *each* group

> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at δ = %s: 8\n",
+ params$pwr.target, params$delta[2]))
Required n1 = n2 = n for Power ≥ 0.85 at δ = 4: 8
> power.t.test(n = c(7, 8), delta = params$delta[2], sd = s_p,
+ sig.level = params$alpha, alternative = "one.sided"
+ )

     Two-sample t test power calculation

              n = 7, 8
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.8326462, 0.8789012
    alternative = one.sided

NOTE: n is number in *each* group

> # 6.3 Visualizations
> # a. Hypothesis Testing (Current n_1 = n_2 = n = 8)
> # Showing the Effect of δ (≡ Δ − Δ₀) on Power
> NTplot(t.test(Cat1, Cat2, alternative = "greater", var.equal = TRUE),
+ mean1 = Delta[2], zaxis = T, power = T)
```

```
> # b. Confidence Interval (Current n_1 = n_2 = n = 8)
> NTplot(t.test(Cat1, Cat2, alternative = "greater", var.equal = TRUE),
+ type = "confidence", zaxis = T)
> # c. Comparison of Sample Sizes (Targeting Required n)
> # Plotting n - 1 vs. required n_1 = n_2 = n = 10 at δ = 4:
> power.t.test(delta = params$delta[2], sd = s_p, sig.level = params$alph,
+ power = params$pwr.target, alternative = "one.sided")

     Two-sample t test power calculation

              n = 7.341304
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = one.sided

NOTE: n is number in *each* group

> NTplot(power.t.test(n = 7, delta = params$delta[2], sd = s_p,
+ sig.level = params$alpha, alternative = "one.sided"), zaxis = T,
+ power = T)
> NTplot(power.t.test(n = 8, delta = params$delta[2], sd = s_p,
+ sig.level = params$alpha, alternative = "one.sided"), zaxis = T,
+ power = T)
> # --- 7. One-Sided (Less) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (LESS) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (LESS) TEST ANALYSIS ===\n")

=== ONE-SIDED (LESS) TEST ANALYSIS ===
> (CritVal_le <- qt(c(0, 1 - params$alpha), df_Case1))
[1]    -Inf 1.76131
> (PVal_le <- pt(t0, df_Case1))
[1] 0.3644568
> (CI_le <- -diff(xbar) + CritVal_le * se_Case1)
[1]     -Inf 1.901027
> t.test(Cat1, Cat2, alternative = "less", var.equal = TRUE)

        Two Sample t-test

data:  Cat1 and Cat2
t = -0.35359, df = 14, p-value = 0.3645
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf 1.901027
sample estimates:
mean of x mean of y
  92.2550   92.7325

> cat(sprintf("Test Statistic (t0): %.3f\nP-Value: %.4f\n", t0, PVal_le))
Test Statistic (t0): -0.354
P-Value: 0.3645
> cat("Confidence Interval:", "(-∞,", CI_le[2], "]\n")
Confidence Interval: (-∞, 1.901027 ]
> # 7.1 Power at δ = -4:
> Pwr.t.test(n_Case1, params$delta[1], s_Case1, params$alpha, "less")
[1] 0.8789012
> power.t.test(n = params$n[1], delta = abs(params$delta[1]), sd = s_p,
+ sig.level = params$alpha, alternative = "one.sided")

     Two-sample t test power calculation

              n = 8
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.8789012
    alternative = one.sided

NOTE: n is number in *each* group

> # 7.2 Sample Size Requirements (n_1 = n_2 = n)
```

```
> power.t.test(delta = abs(params$delta[1]), sd = s_p,
+ sig.level = params$alpha, power = params$pwr.target,
+ alternative = "one.sided")

     Two-sample t test power calculation

              n = 7.341304
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = one.sided

NOTE: n is number in *each* group

> cat(sprintf("Required n1 = n2 = n for Power ≥ %.2f at δ = %s: 8\n",
+ params$pwr.target, params$delta[1]))
Required n1 = n2 = n for Power ≥ 0.85 at δ = -4: 8
> power.t.test(n = c(7, 8), delta = abs(params$delta[1]), sd = s_p,
+ sig.level = params$alph, alternative = "one.sided")

     Two-sample t test power calculation

              n = 7, 8
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.8326462, 0.8789012
    alternative = one.sided

NOTE: n is number in *each* group

> # 7.3 Visualizations
> # a. Hypothesis Testing (Current n_1 = n_2 = n = 8)
> # Showing the Effect of δ (≡ Δ − Δ₀) on Power
> NTplot(t.test(Cat1, Cat2, alternative = "less", var.equal = TRUE),
+ mean1 = Delta[1], zaxis = T, power = T)
> # b. Confidence Interval (Current n_1 = n_2 = n = 8)
> NTplot(t.test(Cat1, Cat2, alternative = "less", var.equal = TRUE),
+ type = "confidence", zaxis = T)
> # c. Comparison of Sample Sizes (Targeting Required n)
> # Plotting n − 1 vs. required n_1 = n_2 = n = 10 at δ = -4, 4:
> power.t.test(delta = abs(params$delta[1]), sd = s_p,
+ sig.level = params$alph, power = params$pwr.target,
+ alternative = "one.sided")

     Two-sample t test power calculation

              n = 7.341304
          delta = 4
             sd = 2.700862
      sig.level = 0.05
          power = 0.85
    alternative = one.sided

NOTE: n is number in *each* group

> NTplot(power.t.test(n = 7, delta = abs(params$delta[1]), sd = s_p,
+ sig.level = params$alph, alternative = "one.sided"), zaxis = T,
+ power = T)
> NTplot(power.t.test(n = 8, delta = abs(params$delta[1]), sd = s_p,
+ sig.level = params$alpha, alternative = "one.sided"), zaxis = T,
+ power = T)
> # Case 1: σ_1² ≠ σ_2²
> # EXAMPLE 5-5 Arsenic in Drinking Water
> # Figure 5-3 Normal probability of the arsenic concentration data from
> # Example 5-5.
> # --- 3*. Data & Parameters ---
> (PHX <- c( 3, 7, 25, 10, 15, 6, 12, 25, 15, 7))
 [1]  3  7 25 10 15  6 12 25 15  7
> (RuralAZ <- c(48, 44, 40, 38, 33, 21, 20, 12, 1, 18))
 [1] 48 44 40 38 33 21 20 12  1 18
> (params <- list(Delta0 = 0, alpha = 0.05,
+ n = c(length(PHX), length(RuralAZ))))
```

```
$Delta0
[1] 0

$alpha
[1] 0.05

$n
[1] 10 10

> # --- 4*. Basic Stats, Normality Test, Box Plots & Variance Test ---
> summary(PHX)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
    3.0     7.0    11.0    12.5    15.0    25.0 
> summary(RuralAZ)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
    1.0    18.5    27.0    27.5    39.5    48.0 
> (xbar <- c(mean(PHX), mean(RuralAZ)))
[1] 12.5 27.5
> (s <- c(sd(PHX), sd(RuralAZ)))
[1]  7.633988 15.349629
> (df <- params$n - 1)
[1] 9 9
> (df_Case2 <- sum(s^2/params$n)^2 / sum((s^2/params$n)^2/(params$n - 1)))
[1] 13.19556
> (se_Case2 <- sqrt(sum(s^2 / params$n)))
[1] 5.421152
> (tstar0 <- (-diff(xbar) - params$Delta0) / se_Case2)
[1] -2.76694
> # Normality Test
> shapiro.test(PHX)

        Shapiro-Wilk normality test

data:  PHX
W = 0.89361, p-value = 0.1861

> shapiro.test(RuralAZ)

        Shapiro-Wilk normality test

data:  RuralAZ
W = 0.95066, p-value = 0.6764

> Conc <- c(PHX, RuralAZ)
> Type <- c(rep(1, params$n[1]), rep(2, params$n[2]))
> Type <- factor(Type, c("1", "2"), c("PHX", "RuralAZ"))
> qqPlot(Conc ~ Type, ylab = "Concentration (ppb)")
> # Comparative Box Plots
> sp(Conc ~ Type, xlab = "Community type", ylab = "Concentration (ppb)")
> # Variance Test
> var.test(PHX, RuralAZ)

        F test to compare two variances

data:  PHX and RuralAZ
F = 0.24735, num df = 9, denom df = 9, p-value = 0.04936
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.06143758 0.99581888
sample estimates:
ratio of variances 
         0.2473473 

> # --- 5*. Two-Sided Test Analysis ---
> # ==========================================================================
> # CASE: TWO-SIDED TEST
> # ==========================================================================
> cat("\n=== TWO-SIDED TEST ANALYSIS ===\n")

=== TWO-SIDED TEST ANALYSIS ===
> (CritVal_2s <- qt(c(params$alpha/2, 1 - params$alpha/2), df_Case2))
[1] -2.157118  2.157118
> (PVal_2s <- 2 * (1 - pt(abs(tstar0), df_Case2)))
[1] 0.01582728
```

```
> (CI_2s <- -diff(xbar) + CritVal_2s * se_Case2)
[1] -26.694067  -3.305933
> t.test(PHX, RuralAZ)

        Welch Two Sample t-test

data:  PHX and RuralAZ
t = -2.7669, df = 13.196, p-value = 0.01583
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -26.694067  -3.305933
sample estimates:
mean of x mean of y
     12.5      27.5

> cat(sprintf("Test Statistic (t*0): %.3f\nP-Value: %.4f\n", tstar0,
+ PVal_2s))
Test Statistic (t*0): -2.767
P-Value: 0.0158
> cat("Confidence Interval:", "[", CI_2s[1], ",", CI_2s[2], "]\n")
Confidence Interval: [ -26.69407 , -3.305933 ]
> # 5*.1 Visualizations
> # a. Hypothesis Testing
> NTplot(t.test(PHX, RuralAZ), zaxis = T)
> # b. Confidence Interval
> NTplot(t.test(PHX, RuralAZ), type = "confidence", zaxis = T)
> # --- 6*. One-Sided (Greater) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (GREATER) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (GREATER) TEST ANALYSIS ===\n")

=== ONE-SIDED (GREATER) TEST ANALYSIS ===
> (CritVal_gr <- qt(c(params$alpha, 1), df_Case2))
[1] -1.768928       Inf
> (PVal_gr <- 1 - pt(tstar0, df_Case2))
[1] 0.9920864
> (CI_gr <- -diff(xbar) + CritVal_gr * se_Case2)
[1] -24.58963       Inf
> t.test(PHX, RuralAZ, alternative = "greater")

        Welch Two Sample t-test

data:  PHX and RuralAZ
t = -2.7669, df = 13.196, p-value = 0.9921
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -24.58963       Inf
sample estimates:
mean of x mean of y
     12.5      27.5

> cat(sprintf("Test Statistic (t*0): %.3f\nP-Value: %.4f\n", tstar0,
+ PVal_gr))
Test Statistic (t*0): -2.767
P-Value: 0.9921
> cat("Confidence Interval:", "[", CI_gr[1], ", ∞)\n")
Confidence Interval: [ -24.58963 , ∞)
> # 6*.1 Visualizations
> # a. Hypothesis Testing
> NTplot(t.test(PHX, RuralAZ, alternative = "greater"), zaxis = T)
> # b. Confidence Interval
> NTplot(t.test(PHX, RuralAZ, alternative = "greater"), type = "confidence",
+ zaxis = T)
> # --- 7*. One-Sided (Less) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (LESS) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (LESS) TEST ANALYSIS ===\n")

=== ONE-SIDED (LESS) TEST ANALYSIS ===
> (CritVal_le <- qt(c(0, 1 - params$alpha), df_Case2))
[1]     -Inf 1.768928
> (PVal_le <- pt(tstar0, df_Case2))
```

```
[1] 0.007913642
> (CI_le <- -diff(xbar) + CritVal_le * se_Case2)
[1]     -Inf -5.41037
> t.test(PHX, RuralAZ, alternative = "less")

        Welch Two Sample t-test

data:  PHX and RuralAZ
t = -2.7669, df = 13.196, p-value = 0.007914
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf -5.41037
sample estimates:
mean of x mean of y
     12.5      27.5

> cat(sprintf("Test Statistic (t*0): %.3f\nP-Value: %.4f\n", tstar0,
+ PVal_le))
Test Statistic (t*0): -2.767
P-Value: 0.0079
> cat("Confidence Interval:", "(-∞,", CI_le[2], "]\n")
Confidence Interval: (-∞, -5.41037 ]
> # 7*.1 Visualizations
> # a. Hypothesis Testing
> NTplot(t.test(PHX, RuralAZ, alternative = "less"), zaxis = T)
> # b. Confidence Interval
> NTplot(t.test(PHX, RuralAZ, alternative = "less"), type = "confidence",
+ zaxis = T)
> # § 5-3.2 Type II Error and Choice of Sample Size
> # Case 1: σ_1² = σ_2² = σ²
> # EXAMPLE 5-6 Chemical Process Yield
> # Refer to EXAMPLE 5-4.
> # § 5-3.3 Confidence Interval on the Difference in Means
> # Case 1: σ_1² = σ_2² = σ²
> # EXAMPLE 5-7 Calcium in Doped Cement
> # Refer to EXAMPLE 5-4.
> # --- 3**. Parameters ---
> (params <- list(Delta0 = 0, alpha = 0.05, n = c(10, 15)))
$Delta0
[1] 0

$alpha
[1] 0.05

$n
[1] 10 15

> # --- 4**. Basic Stats & Data Simulation ---
> (xbar <- c(90.0, 87.0))
[1] 90 87
> (s <- c(5.0, 4.0))
[1] 5 4
> (df <- params$n - 1)
[1]  9 14
> (df_Case1 <- sum(df))
[1] 23
> (s_p <- sqrt(sum(df * s^2) / df_Case1))
[1] 4.418341
> (se_Case1 <- s_p * sqrt(sum(1 / params$n)))
[1] 1.80378
> (t0 <- (-diff(xbar) - params$Delta0) / se_Case1)
[1] 1.663174
> set.seed(123)
> Std_raw <- rnorm(params$n[1])
> # Rescale to have sample mean xbar[1] and sample standard deviation s[1]
> Std <- xbar[1] + s[1] * (Std_raw - mean(Std_raw)) / sd(Std_raw)
> Lead_raw <- rnorm(params$n[2])
> # Rescale to have sample mean xbar[2] and sample standard deviation s[2]
> Lead <- xbar[2] + s[2] * (Lead_raw - mean(Lead_raw)) / sd(Lead_raw)
> # --- 5**. Two-Sided Test Analysis ---
> # ==========================================================================
> # CASE: TWO-SIDED TEST
> # ==========================================================================
> cat("\n=== TWO-SIDED TEST ANALYSIS ===\n")
```

```
=== TWO-SIDED TEST ANALYSIS ===
> (CritVal_2s <- qt(c(params$alpha/2, 1 - params$alpha/2), df_Case1))
[1] -2.068658  2.068658
> (PVal_2s <- 2 * (1 - pt(abs(t0), df_Case1)))
[1] 0.1098463
> (CI_2s <- -diff(xbar) + CritVal_2s * se_Case1)
[1] -0.7314037  6.7314037
> t.test(Std, Lead, var.equal = TRUE)

        Two Sample t-test

data:  Std and Lead
t = 1.6632, df = 23, p-value = 0.1098
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.7314037  6.7314037
sample estimates:
mean of x mean of y
       90        87

> cat(sprintf("Test Statistic (t0): %.3f\nP-Value: %.4f\n", t0, PVal_2s))
Test Statistic (t0): 1.663
P-Value: 0.1098
> cat("Confidence Interval:", "[", CI_2s[1], ",", CI_2s[2], "]\n")
Confidence Interval: [ -0.7314037 , 6.731404 ]
> # 5**.1 Visualizations
> # a. Hypothesis Testing
> NTplot(t.test(Std, Lead, var.equal = TRUE), zaxis = T)
> # b. Confidence Interval (Current n_1 = n_2 = n = 8)
> NTplot(t.test(Std, Lead, var.equal = TRUE), type = "confidence", zaxis = T)
> # --- 6**. One-Sided (Greater) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (GREATER) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (GREATER) TEST ANALYSIS ===\n")

=== ONE-SIDED (GREATER) TEST ANALYSIS ===
> (CritVal_gr <- qt(c(params$alpha, 1), df_Case1))
[1] -1.713872       Inf
> (PVal_gr <- 1 - pt(t0, df_Case1))
[1] 0.05492316
> (CI_gr <- -diff(xbar) + CritVal_gr * se_Case1)
[1] -0.09144761         Inf
> t.test(Std, Lead, alternative = "greater", var.equal = TRUE)

        Two Sample t-test

data:  Std and Lead
t = 1.6632, df = 23, p-value = 0.05492
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -0.09144761         Inf
sample estimates:
mean of x mean of y
       90        87

> cat(sprintf("Test Statistic (t0): %.3f\nP-Value: %.4f\n", t0, PVal_gr))
Test Statistic (t0): 1.663
P-Value: 0.0549
> cat("Confidence Interval:", "[", CI_gr[1], ", ∞)\n")
Confidence Interval: [ -0.09144761 , ∞)
> # 6**.1 Visualizations
> # a. Hypothesis Testing
> NTplot(t.test(Std, Lead, alternative = "greater", var.equal = TRUE),
+ zaxis = T)
> # b. Confidence Interval
> NTplot(t.test(Std, Lead, alternative = "greater", var.equal = TRUE),
+ type = "confidence", zaxis = T)
> # --- 7**. One-Sided (Less) Test Analysis ---
> # ==========================================================================
> # CASE: ONE-SIDED (LESS) TEST
> # ==========================================================================
> cat("\n=== ONE-SIDED (LESS) TEST ANALYSIS ===\n")
```

```
=== ONE-SIDED (LESS) TEST ANALYSIS ===
> (CritVal_le <- qt(c(0, 1 - params$alpha), df_Case1))
[1]     -Inf 1.713872
> (PVal_le <- pt(t0, df_Case1))
[1] 0.9450768
> (CI_le <- -diff(xbar) + CritVal_le * se_Case1)
[1]     -Inf 6.091448
> t.test(Std, Lead, alternative = "less", var.equal = TRUE)

        Two Sample t-test

data:  Std and Lead
t = 1.6632, df = 23, p-value = 0.9451
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf 6.091448
sample estimates:
mean of x mean of y
       90        87

> cat(sprintf("Test Statistic (t0): %.3f\nP-Value: %.4f\n", t0, PVal_le))
Test Statistic (t0): 1.663
P-Value: 0.9451
> cat("Confidence Interval:", "(-∞,", CI_le[2], "]\n")
Confidence Interval: (-∞, 6.091448 ]
> # 7**.1. Visualizations
> # a. Hypothesis Testing
> NTplot(t.test(Std, Lead, alternative = "less", var.equal = TRUE),
+ zaxis = T)
> # b. Confidence Interval
> NTplot(t.test(Std, Lead, alternative = "less", var.equal = TRUE),
+ type = "confidence", zaxis = T)
> # EXERCISES FOR § 5-3
> # § 5-4 The Paired t-Test
> # Refer to § 4-5.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(car, HH)
> # EXAMPLE 5-8 Shear Strengh of Steel Girders
> # Table 5-3 Strength Predictions for Nine Steel Plate Girders (Predicted
> # Load/Observed Load)
> Karlsruhe <- c(1.186, 1.151, 1.322, 1.339, 1.200, 1.402, 1.365, 1.537,
+ 1.559)
> Lehigh <- c(1.061, 0.992, 1.063, 1.062, 1.065, 1.178, 1.037, 1.086, 1.052)
> (KL.diff <- Karlsruhe - Lehigh)
[1] 0.125 0.159 0.259 0.277 0.135 0.224 0.328 0.451 0.507
> ObsNo. <- 1:length(KL.diff); ShearStr <- data.frame(KL.diff, ObsNo.)
> summary(ShearStr); spm(ShearStr)
    KL.diff           ObsNo.
 Min.   :0.1250   Min.   :1
 1st Qu.:0.1590   1st Qu.:3
 Median :0.2590   Median :5
 Mean   :0.2739   Mean   :5
 3rd Qu.:0.3280   3rd Qu.:7
 Max.   :0.5070   Max.   :9
> shapiro.test(KL.diff); qqp(KL.diff)

        Shapiro-Wilk normality test

data:  KL.diff
W = 0.91678, p-value = 0.3663

[1] 9 8
> (dbar <- mean(KL.diff)); (s <- sd(KL.diff)); (Delta0 <- 0)
[1] 0.2738889
[1] 0.1350994
[1] 0
> (sig.lv <- .05); (n <- length(KL.diff)); (df <- n - 1); (se <- s/sqrt(n))
[1] 0.05
[1] 9
[1] 8
[1] 0.04503315
> (t0 <- (dbar - Delta0)/se)
```

```
[1] 6.081939
> # alt = "two.sided":
> t.test(Karlsruhe, Lehigh, paired = TRUE)

        Paired t-test

data:  Karlsruhe and Lehigh
t = 6.0819, df = 8, p-value = 0.0002953
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.1700423 0.3777355
sample estimates:
mean difference
      0.2738889

> (CritVal <- qt(c(sig.lv/2, 1 - sig.lv/2), df))
[1] -2.306004  2.306004
> (PVal <- 2*(1 - pt(abs(t0), df)))
[1] 0.0002952955
> (CI <- dbar + CritVal*se)
[1] 0.1700423 0.3777355
> NTplot(t.test(Karlsruhe, Lehigh, paired = TRUE), zaxis = T, power = T)
> NTplot(t.test(Karlsruhe, Lehigh, paired = TRUE), "c", zaxis = T)
> # alt = "gr":
> t.test(Karlsruhe, Lehigh, alternative = "greater", paired = TRUE)

        Paired t-test

data:  Karlsruhe and Lehigh
t = 6.0819, df = 8, p-value = 0.0001476
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 0.1901476       Inf
sample estimates:
mean difference
      0.2738889

> (CritVal <- qt(1 - sig.lv, df))
[1] 1.859548
> (PVal <- 1 - pt(t0, df))
[1] 0.0001476477
> (CI <- c(dbar - CritVal*se, Inf))
[1] 0.1901476       Inf
> NTplot(t.test(Karlsruhe, Lehigh, alternative = "greater", paired = TRUE),
+ zaxis = T, power = T)
> NTplot(t.test(Karlsruhe, Lehigh, alternative = "greater", paired = TRUE),
+ "c", zaxis = T)
> # alt = "less":
> t.test(Karlsruhe, Lehigh, alternative = "less", paired = TRUE)

        Paired t-test

data:  Karlsruhe and Lehigh
t = 6.0819, df = 8, p-value = 0.9999
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf 0.3576302
sample estimates:
mean difference
      0.2738889

> (CritVal <- qt(sig.lv, df))
[1] -1.859548
> (PVal <- pt(t0, df))
[1] 0.9998524
> (CI <- c(-Inf, dbar - CritVal*se))
[1]      -Inf 0.3576302
> NTplot(t.test(Karlsruhe, Lehigh, alternative = "less", paired = TRUE),
+ zaxis = T, power = T)
> NTplot(t.test(Karlsruhe, Lehigh, alternative = "less", paired = TRUE), "c",
+ zaxis = T)
> # EXAMPLE 5-9 Parallel Parking
> # Refer to EXAMPLE 5-8.
> # Table 5-4 Time in Seconds to Parallel Park Two Automobiles
```

```
> Auto1 <- c(37.0, 25.8, 16.2, 24.2, 22.0, 33.4, 23.8, 58.2, 33.6, 24.4,
+ 23.4, 21.2, 36.2, 29.8)
> Auto2 <- c(17.8, 20.2, 16.8, 41.4, 21.4, 38.4, 16.8, 32.2, 27.8, 23.2,
+ 29.6, 20.6, 32.2, 53.8)
> (Auto.diff <- Auto1 - Auto2)
 [1]  19.2   5.6  -0.6 -17.2   0.6  -5.0   7.0  26.0   5.8   1.2  -6.2   0.6
[13]   4.0 -24.0
> ObsNo. <- 1:length(Auto.diff); ParP <- data.frame(Auto.diff, ObsNo.)
> summary(ParP); spm(ParP)
   Auto.diff           ObsNo.
 Min.   :-24.000   Min.   : 1.00
 1st Qu.: -3.900   1st Qu.: 4.25
 Median :  0.900   Median : 7.50
 Mean   :  1.214   Mean   : 7.50
 3rd Qu.:  5.750   3rd Qu.:10.75
 Max.   : 26.000   Max.   :14.00
> shapiro.test(Auto.diff); qqp(Auto.diff)

        Shapiro-Wilk normality test

data:  Auto.diff
W = 0.94912, p-value = 0.5472

[1] 14  8
> (dbar <- mean(Auto.diff)); (s <- sd(Auto.diff)); (Delta0 <- 0)
[1] 1.214286
[1] 12.68493
[1] 0
> (sig.lv <- .1); (n <- length(Auto.diff)); (df <- n - 1); (se <- s/sqrt(n))
[1] 0.1
[1] 14
[1] 13
[1] 3.39019
> # alt = "two.sided":
> t.test(Auto1, Auto2, paired = TRUE, conf.level = 0.9)

        Paired t-test

data:  Auto1 and Auto2
t = 0.35818, df = 13, p-value = 0.726
alternative hypothesis: true mean difference is not equal to 0
90 percent confidence interval:
 -4.789516  7.218087
sample estimates:
mean difference
       1.214286

> (CritVal <- qt(c(sig.lv/2, 1 - sig.lv/2), df))
[1] -1.770933  1.770933
> (CI <- dbar + CritVal*se)
[1] -4.789516  7.218087
> NTplot(t.test(Auto1, Auto2, paired = TRUE, conf.level = 0.9), zaxis = T,
+ power = T)
> NTplot(t.test(Auto1, Auto2, paired = TRUE, conf.level = 0.9), "c",
+ zaxis = T)
> # alt = "gr":
> NTplot(t.test(Auto1, Auto2, alternative = "greater", paired = TRUE,
+ conf.level = 0.9), zaxis = T, power = T)
> NTplot(t.test(Auto1, Auto2, alternative = "greater", paired = TRUE,
+ conf.level = 0.9), "c", zaxis = T)
> # alt = "less":
> NTplot(t.test(Auto1, Auto2, alternative = "less", paired = TRUE,
+ conf.level = 0.9), zaxis = T, power = T)
> NTplot(t.test(Auto1, Auto2, alternative = "less", paired = TRUE,
+ conf.level = 0.9), "c", zaxis = T)
> # EXERCISES FOR § 5-4
> # § 5-5 Inference on the Ratio of Variances of Two Normal Populations
> # Refer to § 4-6.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(car, HH)
> # § 5-5.1 Hypothesis Testing on the Ratio of Two Variances
> # Figure 5-4 Probability density functions of two F distributions.
> old.omd <- par(omd = c(.07, .88, 0, 1))
```

```
> F.setup(5, 15, xlim.in = c(0, qf(.95, 5, 5)),
+ main.in = expression("F density:" ~~ nu[1]=="5" ~~ nu[2]=="5, 15"))
> F.curve(5, 5); F.curve(5, 15, col = 2)
> F.setup(xlim.in = c(0, qf(.95, 2, 2)), ylim.in = c(0, 1.5), main.in =
+ expression("F density:" ~~ nu[1]=="1, 2, …, 10" ~~ nu[2]==nu[1]))
> for (i in 1:10) F.curve(i, i, f = seq(0, par()$usr[2], length = 1001),
+ col = i)
> F.setup(20, 20, xlim.in = c(0, qf(.95, 11, 11)), main.in =
+ expression("F density:" ~~ nu[1]=="11, 12, …, 20" ~~ nu[2]==nu[1]))
> for (i in 11:20) F.curve(i, i, col = i - 10)
> F.setup(30, 30, xlim.in = c(0, qf(.95, 21, 21)), main.in =
+ expression("F density:" ~~ nu[1]=="21, 22, …, 30" ~~ nu[2]==nu[1]))
> for (i in 21:30) F.curve(i, i, col = i - 20)
> F.setup(120, 120, xlim.in = c(0, qf(.95, 10, 10)), main.in =
+ expression("F density:" ~~ nu[1]=="10, 20, …, 120" ~~ nu[2]==nu[1]))
> for (i in 10*(1:12)) F.curve(i, i, col = i/10)
> par(old.omd)
> # Figure 5-5 Upper and lower percentage points of the F distribution.
> # Figure 5-6 The F distribution for the test of H0: sigma1^2 = sigma2^2
> # with critical region values for (a) H1:sigma1^2 not = sigma2^2, (b) H1:
> # sigma1^2 > sigma2^2, and (c) H1: sigma1^2 < sigma2^2.
> Pwr.var.test.t <- function (n, rto0, rto1, sig.lv) {
+ df <- n - 1
+ 1 - pf(qf(1 - sig.lv/2, df[1], df[2])*rto0/rto1, df[1], df[2]) +
+ pf(qf(sig.lv/2, df[1], df[2])*rto0/rto1, df[1], df[2])}
> Pwr.var.test.g <- function (n, rto0, rto1, sig.lv) {
+ df <- n - 1; 1 - pf(qf(1 - sig.lv, df[1], df[2])*rto0/rto1, df[1], df[2])}
> Pwr.var.test.l <- function (n, rto0, rto1, sig.lv) {
+ df <- n - 1; pf(qf(sig.lv, df[1], df[2])*rto0/rto1, df[1], df[2])}
> # EXAMPLE 5-10 Etching Wafers
> (n <- c(16, 16)); (s <- c(1.96, 2.13)); (rto0 <- 1); (sig.lv <- .05)
[1] 16 16
[1] 1.96 2.13
[1] 1
[1] 0.05
> (f0 <- (s[1]^2/s[2]^2)/rto0); (df <- n - 1); (rto1 <- c(2^2, .5^2))
[1] 0.8467456
[1] 15 15
[1] 4.00 0.25
> x1.0 <- rnorm(n[1]); x1 <- s[1]*(x1.0 - mean(x1.0))/sd(x1.0)
> # x1 is a simulated normal data set with sample size n[1] and sample
> # standard deviation s[1].
> length(x1); sd(x1)
[1] 16
[1] 1.96
> x2.0 <- rnorm(n[2]); x2 <- s[2]*(x2.0 - mean(x2.0))/sd(x2.0)
> # x2 is a simulated normal data set with sample size n[2] and sample
> # standard deviation s[2].
> length(x2); sd(x2)
[1] 16
[1] 2.13
> # alt = "two.sided":
> var.test(x1, x2)

        F test to compare two variances

data:  x1 and x2
F = 0.84675, num df = 15, denom df = 15, p-value = 0.7515
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.2958484 2.4234642
sample estimates:
ratio of variances
         0.8467456

> (CritVal <- qf(c(sig.lv/2, 1 - sig.lv/2), df[1], df[2]))
[1] 0.3493947 2.8620925
> (PVal <- 2*min(pf(f0, df[1], df[2]), 1 - pf(f0, df[1], df[2])))
[1] 0.7515129
> Pwr.var.test.t(n, rto0, rto1, sig.lv)
[1] 0.7376452 0.7376452
> (CI <- f0/qf(c(1 - sig.lv/2, sig.lv/2), df[1], df[2]))
[1] 0.2958484 2.4234642
> (qf <- range(c(f0, qf(1 - pf(f0, df[1], df[2]), df[1], df[2]))))
```

```
[1] 0.8467456 1.1809923
> old.omd <- par(omd = c(.07, .88, .05, 1))
> F.setup(df[1], df[2])
> F.curve(df[1], df[2], 0, F, c(sig.lv/2, sig.lv/2), col = 'blue')
> F.observed(qf[1], "green", df[1], df[2], 0, F, "f", "left")
> F.observed(qf[2], "green", df[1], df[2])
> axis(3, 2.5, expression(n[1]==nu[1] + 1 ~~ n[2]==nu[2] + 1 ~~ s[1]==1.96 ~~
+ s[2]==2.13 ~~ f[0]==s[1]^2/s[2]^2), T, 1.5, lwd = 0)
> axis(3, 2.5, expression(LCL ~~ UCL ==f[0]/f[nu[1]~nu[2]]), T, -2.5,
+ lwd = 0)
> axis(1, 5.2, round(PVal, 4), T, 3.5, lwd = 0)
> axis(1, -.2, expression(f[0]/f[nu[1]~nu[2]]), T, 5, lwd = 0)
> axis(1, qf(c(sig.lv/2, 1 - sig.lv/2), df[1], df[2]),
+ round(f0/qf(c(sig.lv/2, 1 - sig.lv/2), df[1], df[2]), 4), T, 5, lwd = 0)
> par(old.omd)
> # alt = "gr":
> var.test(x1, x2, rto0, "g")

        F test to compare two variances

data:  x1 and x2
F = 0.84675, num df = 15, denom df = 15, p-value = 0.6242
alternative hypothesis: true ratio of variances is greater than 1
95 percent confidence interval:
 0.3523046       Inf
sample estimates:
ratio of variances
         0.8467456

> (CritVal <- qf(1 - sig.lv, df[1], df[2]))
[1] 2.403447
> (PVal <- 1 - pf(f0, df[1], df[2]))
[1] 0.6242436
> Pwr.var.test.g(n, rto0, rto1[1], sig.lv)
[1] 0.832742
> (CI <- c(f0/qf(1 - sig.lv, df[1], df[2]), Inf))
[1] 0.3523046       Inf
> old.omd <- par(omd = c(.07, .88, .05, 1))
> F.setup(df[1], df[2])
> F.curve(df[1], df[2], 0, F, sig.lv, col = 'blue')
> F.observed(f0, "green", df[1], df[2])
> axis(3, 2.5, expression(n[1]==nu[1] + 1 ~~ n[2]==nu[2] + 1 ~~ s[1]==1.96 ~~
+ s[2]==2.13 ~~ f[0]==s[1]^2/s[2]^2), T, 1.5, lwd = 0)
> axis(3, 2.5, expression(LCL ==f[0]/f[nu[1]~nu[2]]), T, -2.5, lwd = 0)
> axis(1, 5.2, round(PVal, 4), T, 3.5, lwd = 0)
> axis(1, -.2, expression(f[0]/f[nu[1]~nu[2]]), T, 5, lwd = 0)
> axis(1, qf(1 - sig.lv, df[1], df[2]),
+ round(f0/qf(1 - sig.lv, df[1], df[2]), 4), T, 5, lwd = 0)
> par(old.omd)
> # alt = "less":
> var.test(x1, x2, rto0, "l")

        F test to compare two variances

data:  x1 and x2
F = 0.84675, num df = 15, denom df = 15, p-value = 0.3758
alternative hypothesis: true ratio of variances is less than 1
95 percent confidence interval:
 0.000000 2.035108
sample estimates:
ratio of variances
         0.8467456

> (CritVal <- qf(sig.lv, df[1], df[2]))
[1] 0.4160691
> (PVal <- pf(f0, df[1], df[2]))
[1] 0.3757564
> Pwr.var.test.l(n, rto0, rto1[2], sig.lv)
[1] 0.832742
> (CI <- c(0, f0/qf(sig.lv, df[1], df[2])))
[1] 0.000000 2.035108
> old.omd <- par(omd = c(.07, .88, .05, 1))
> F.setup(df[1], df[2])
> F.curve(df[1], df[2], 0, F, 1 - sig.lv, shade = "l", col = 'blue')
```

```
> F.observed(f0, "green", df[1], df[2], 0, F, "f", "left")
> axis(3, 2.5, expression(n[1]==nu[1] + 1 ~~ n[2]==nu[2] + 1 ~~ s[1]==1.96 ~~
+ s[2]==2.13 ~~ f[0]==s[1]^2/s[2]^2), T, 1.5, lwd = 0)
> axis(3, 2.5, expression(UCL ==f[0]/f[nu[1]~nu[2]]), T, -2.5, lwd = 0)
> axis(1, 5.2, round(PVal, 4), T, 3.5, lwd = 0)
> axis(1, -.2, expression(f[0]/f[nu[1]~nu[2]]), T, 5, lwd = 0)
> axis(1, qf(sig.lv, df[1], df[2]), round(f0/qf(sig.lv, df[1], df[2]), 4), T,
+ 5, lwd = 0)
> par(old.omd)
> # § 5-5.2 Confidence Interval on the Ratio of Two Variances
> # EXAMPLE 5-11 Surface Finish
> # Refer to EXAMPLE 5-10.
> (n <- c(11, 16)); (s <- c(5.1, 4.7)); (rto0 <- 1); (sig.lv <- .1)
[1] 11 16
[1] 5.1 4.7
[1] 1
[1] 0.1
> (f0 <- (s[1]^2/s[2]^2)/rto0); (df <- n - 1)
[1] 1.177456
[1] 10 15
> x1.0 <- rnorm(n[1]); x1 <- s[1]*(x1.0 - mean(x1.0))/sd(x1.0)
> # x1 is a simulated normal data set with sample size n[1] and sample
> # standard deviation s[1].
> length(x1); sd(x1)
[1] 11
[1] 5.1
> x2.0 <- rnorm(n[2]); x2 <- s[2]*(x2.0 - mean(x2.0))/sd(x2.0)
> # x2 is a simulated normal data set with sample size n[2] and sample
> # standard deviation s[2].
> length(x2); sd(x2)
[1] 16
[1] 4.7
> # alt = "two.sided":
> var.test(x1, x2, rto0, "t", 1 - sig.lv)

        F test to compare two variances

data:  x1 and x2
F = 1.1775, num df = 10, denom df = 15, p-value = 0.7509
alternative hypothesis: true ratio of variances is not equal to 1
90 percent confidence interval:
 0.4628876 3.3498814
sample estimates:
ratio of variances
          1.177456

> (CritVal <- qf(c(sig.lv/2, 1 - sig.lv/2), df[1], df[2]))
[1] 0.3514918 2.5437185
> (PVal <- 2*min(pf(f0, df[1], df[2]), 1 - pf(f0, df[1], df[2])))
[1] 0.7508517
> (CI <- f0/qf(c(1 - sig.lv/2, sig.lv/2), df[1], df[2]))
[1] 0.4628876 3.3498814
> (qf <- range(c(f0, qf(1 - pf(f0, df[1], df[2]), df[1], df[2]))))
[1] 0.8092403 1.1774559
> old.omd <- par(omd = c(.07, .88, .05, 1))
> F.setup(df[1], df[2])
> F.curve(df[1], df[2], 0, F, c(sig.lv/2, sig.lv/2), col = 'blue')
> F.observed(qf[1], "green", df[1], df[2], 0, F, "f", "left")
> F.observed(qf[2], "green", df[1], df[2])
> axis(3, 2.5, expression(n[1]==nu[1] + 1 ~~ n[2]==nu[2] + 1 ~~ s[1]==1.96 ~~
+ s[2]==2.13 ~~ f[0]==s[1]^2/s[2]^2), T, 1.5, lwd = 0)
> axis(3, 2.5, expression(LCL ~~ UCL ==f[0]/f[nu[1]~nu[2]]), T, -2.5,
+ lwd = 0)
> axis(1, 5.2, round(PVal, 4), T, 3.5, lwd = 0)
> axis(1, -.2, expression(f[0]/f[nu[1]~nu[2]]), T, 5, lwd = 0)
> axis(1, qf(c(sig.lv/2, 1 - sig.lv/2), df[1], df[2]),
+ round(f0/qf(c(sig.lv/2, 1 - sig.lv/2), df[1], df[2]), 4), T, 5, lwd = 0)
> par(old.omd)
> # EXERCISES FOR § 5-5
> # § 5-6 Inference on Two Population Proportions
> # Refer to § 4-7.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(car, HH)
```

```
> # § 5-6.1 Hypothesis Testing on the Equality of Two Binomial Proportions
> # EXAMPLE 5-12 Interocular Lenses
> (n <- c(300, 300)); (x <- c(253, 196)); (Delta0 <- 0); (aprx.lv <- .05)
[1] 300 300
[1] 253 196
[1] 0
[1] 0.05
> (phat <- x/n); (qhat <- 1 - phat); (phat.diff <- -diff(phat))
[1] 0.8433333 0.6533333
[1] 0.1566667 0.3466667
[1] 0.19
> (phat.p <- sum(x)/sum(n)); (qhat.p <- 1 - phat.p)
[1] 0.7483333
[1] 0.2516667
> (se.p <- sqrt(phat.p*qhat.p*sum(1/n)))
[1] 0.03543356
> (z0 <- phat.diff/se.p)
[1] 5.362148
> (n0 <- mean(n)); (sd.p <- sqrt(n0)*se.p)
[1] 300
[1] 0.6137272
> (sehat <- sqrt(sum(phat*qhat/n))); (sdhat <- sqrt(n0)*sehat)
[1] 0.03457413
[1] 0.5988415
> (ntilde <- n + 2); (xtilde <- x + 1)
[1] 302 302
[1] 254 197
> (ptilde <- xtilde/ntilde); (qtilde <- 1 - ptilde)
[1] 0.8410596 0.6523179
[1] 0.1589404 0.3476821
> (ptilde.diff <- -diff(ptilde))
[1] 0.1887417
> (setilde <- sqrt(sum(ptilde*qtilde/ntilde)))
[1] 0.03454902
> (n0tilde <- mean(ntilde)); (sdtilde <- sqrt(n0tilde)*setilde)
[1] 302
[1] 0.6003979
> # alt = "two.sided":
> prop.test(x, n, NULL, "t", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  x out of n
X-squared = 28.753, df = 1, p-value = 8.224e-08
alternative hypothesis: two.sided
95 percent confidence interval:
 0.122236 0.257764
sample estimates:
   prop 1    prop 2
0.8433333 0.6533333

> (CritVal <- qnorm(c(aprx.lv/2, 1 - aprx.lv/2))) # Aprx
[1] -1.959964  1.959964
> (PVal <- 2*(1 - pnorm(abs(z0)))) # Aprx
[1] 8.223789e-08
> (AsymptoticCI <- phat.diff + CritVal*sehat) # Aprx
[1] 0.122236 0.257764
> prop.test(xtilde, ntilde, NULL, "t", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  xtilde out of ntilde
X-squared = 28.439, df = 1, p-value = 9.668e-08
alternative hypothesis: two.sided
95 percent confidence interval:
 0.1210269 0.2564565
sample estimates:
   prop 1    prop 2
0.8410596 0.6523179

> (AgrestiCaffoCI <- ptilde.diff + CritVal*setilde) # Aprx
[1] 0.1210269 0.2564565
> normal.and.t.dist(Delta0, NA, phat.diff, sd.p, n0, NA, aprx.lv/2,
+ aprx.lv/2, F, T, T) # Aprx
```

```
> normal.and.t.dist(NA, NA, phat.diff, sdhat, n0, NA, aprx.lv/2, aprx.lv/2,
+ F, F, T, T, "c") # Aprx
> normal.and.t.dist(NA, NA, ptilde.diff, sdtilde, n0tilde, NA, aprx.lv/2,
+ aprx.lv/2, F, F, T, T, "c") # Aprx
> # alt = "gr":
> prop.test(x, n, NULL, "g", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  x out of n
X-squared = 28.753, df = 1, p-value = 4.112e-08
alternative hypothesis: greater
95 percent confidence interval:
 0.1331306 1.0000000
sample estimates:
   prop 1    prop 2
0.8433333 0.6533333

> (CritVal <- qnorm(1 - aprx.lv)) # Aprx
[1] 1.644854
> (PVal <- 1 - pnorm(z0)) # Aprx
[1] 4.111895e-08
> (AsymptoticCI <- c(phat.diff - CritVal*sehat, 1)) # Aprx
[1] 0.1331306 1.0000000
> prop.test(xtilde, ntilde, NULL, "g", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  xtilde out of ntilde
X-squared = 28.439, df = 1, p-value = 4.834e-08
alternative hypothesis: greater
95 percent confidence interval:
 0.1319136 1.0000000
sample estimates:
   prop 1    prop 2
0.8410596 0.6523179

> (AgrestiCaffoCI <- c(ptilde.diff - CritVal*setilde, 1)) # Aprx
[1] 0.1319136 1.0000000
> normal.and.t.dist(Delta0, NA, phat.diff, sd.p, n0, NA, aprx.lv, aprx.lv, F,
+ T) # Aprx
> normal.and.t.dist(NA, NA, phat.diff, sdhat, n0, NA, aprx.lv, aprx.lv, F, F,
+ T, F, "c") # Aprx
> normal.and.t.dist(NA, NA, ptilde.diff, sdtilde, n0tilde, NA, aprx.lv,
+ aprx.lv, F, F, T, F, "c") # Aprx
> # alt = "less":
> prop.test(x, n, NULL, "l", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  x out of n
X-squared = 28.753, df = 1, p-value = 1
alternative hypothesis: less
95 percent confidence interval:
 -1.0000000  0.2468694
sample estimates:
   prop 1    prop 2
0.8433333 0.6533333

> (CritVal <- qnorm(aprx.lv)) # Aprx
[1] -1.644854
> (PVal <- pnorm(z0)) # Aprx
[1] 1
> (AsymptoticCI <- c(-1, phat.diff - CritVal*sehat)) # Aprx
[1] -1.0000000  0.2468694
> prop.test(xtilde, ntilde, NULL, "l", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  xtilde out of ntilde
X-squared = 28.439, df = 1, p-value = 1
alternative hypothesis: less
95 percent confidence interval:
 -1.0000000  0.2455698
```

```
sample estimates:
   prop 1    prop 2
0.8410596 0.6523179

> (AgrestiCaffoCI <- c(-1, ptilde.diff - CritVal*setilde)) # Aprx
[1] -1.0000000  0.2455698
> normal.and.t.dist(Delta0, NA, phat.diff, sd.p, n0, NA, aprx.lv, aprx.lv, F,
+ T, T, F) # Aprx
> normal.and.t.dist(NA, NA, phat.diff, sdhat, n0, NA, aprx.lv, aprx.lv, F, F,
+ F, T, "c") # Aprx
> normal.and.t.dist(NA, NA, ptilde.diff, sdtilde, n0tilde, NA, aprx.lv,
+ aprx.lv, F, F, F, T, "c") # Aprx
> # § 5-6.2 Type II Error and Choice of Sample Size
> AprxPwr.prop.test <- function (n, p, aprx.lv, alt) {
+ Delta <- -diff(p); pbar <- sum(n*p)/sum(n)
+ se0 <- sqrt(pbar*(1 - pbar)*sum(1/n)); se1 <- sqrt(sum(p*(1 - p)/n))
+ m <- Delta/se0; sd <- se1/se0
+ if (alt == 2) 1 - pnorm(qnorm(1 - aprx.lv/2), m, sd) +
+ pnorm(qnorm(aprx.lv/2), m, sd)
+ else 1 - pnorm(qnorm(1 - aprx.lv), abs(m), sd)}
> AprxSplSz.prop.test <- function (p, aprx.lv, pwr.at.least, alt)
+ ceiling((qnorm(1 - aprx.lv/alt)*sqrt(sum(p)*sum(1 - p)/2) +
+ qnorm(pwr.at.least)*sqrt(sum(p*(1 - p))))^2/diff(p)^2)
> # Refer to EXAMPLE 5-12.
> (n <- c(300, 300)); (Delta0 <- 0); (aprx.lv <- .05); (n0 <- mean(n))
[1] 300 300
[1] 0
[1] 0.05
[1] 300
> (pwr.at.least <- .9)
[1] 0.9
> # alt = "two.sided":
> (p <- c(.8, .7)); (q <- 1 - p); (Delta <- -diff(p))
[1] 0.8 0.7
[1] 0.2 0.3
[1] 0.1
> (pbar <- sum(n*p)/sum(n)); (qbar <- 1 - pbar)
[1] 0.75
[1] 0.25
> (se0 <- sqrt(pbar*qbar*sum(1/n))); (se <- sqrt(sum(p*q/n)))
[1] 0.03535534
[1] 0.03511885
> AprxPwr.prop.test(n, p, aprx.lv, 2)
[1] 0.8090264
> power.prop.test(n0, p[1], p[2], aprx.lv, NULL, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 300
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.8090264
    alternative = two.sided

NOTE: n is number in *each* group

> (n.2 <- AprxSplSz.prop.test(p, aprx.lv, pwr.at.least, 2))
[1] 392
> power.prop.test(NULL, p[1], p[2], aprx.lv, pwr.at.least, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 391.947
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.9
    alternative = two.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.2 - 1, n.2 - 1), p, aprx.lv, 2) # Aprx
[1] 0.8993078
```

```
> power.prop.test(n.2 - 1, p[1], p[2], aprx.lv, NULL, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 391
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.8993078
    alternative = two.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.2, n.2), p, aprx.lv, 2)
[1] 0.9000387
> power.prop.test(n.2, p[1], p[2], aprx.lv, NULL, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 392
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.9000387
    alternative = two.sided

NOTE: n is number in *each* group

> (p <- c(.7, .8)); (q <- 1 - p); (Delta <- -diff(p))
[1] 0.7 0.8
[1] 0.3 0.2
[1] -0.1
> (pbar <- sum(n*p)/sum(n)); (qbar <- 1 - pbar)
[1] 0.75
[1] 0.25
> (se0 <- sqrt(pbar*qbar*sum(1/n))); (se <- sqrt(sum(p*q/n)))
[1] 0.03535534
[1] 0.03511885
> AprxPwr.prop.test(n, p, aprx.lv, 2)
[1] 0.8090264
> power.prop.test(n0, p[1], p[2], aprx.lv, NULL, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 300
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.8090264
    alternative = two.sided

NOTE: n is number in *each* group

> (n.2 <- AprxSplSz.prop.test(p, aprx.lv, pwr.at.least, 2))
[1] 392
> power.prop.test(NULL, p[1], p[2], aprx.lv, pwr.at.least, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 391.947
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.9
    alternative = two.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.2 - 1, n.2 - 1), p, aprx.lv, 2) # Aprx
[1] 0.8993078
> power.prop.test(n.2 - 1, p[1], p[2], aprx.lv, NULL, "t", T) # Aprx

     Two-sample comparison of proportions power calculation
```

```
              n = 391
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.8993078
    alternative = two.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.2, n.2), p, aprx.lv, 2)
[1] 0.9000387
> power.prop.test(n.2, p[1], p[2], aprx.lv, NULL, "t", T) # Aprx

     Two-sample comparison of proportions power calculation

              n = 392
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.9000387
    alternative = two.sided

NOTE: n is number in *each* group

> # alt = "gr":
> (p <- c(.8, .7)); (q <- 1 - p); (Delta <- -diff(p))
[1] 0.8 0.7
[1] 0.2 0.3
[1] 0.1
> (pbar <- sum(n*p)/sum(n)); (qbar <- 1 - pbar)
[1] 0.75
[1] 0.25
> (se0 <- sqrt(pbar*qbar*sum(1/n))); (se <- sqrt(sum(p*q/n)))
[1] 0.03535534
[1] 0.03511885
> AprxPwr.prop.test(n, p, aprx.lv, 1)
[1] 0.8832799
> power.prop.test(n0, p[1], p[2], aprx.lv, NULL, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 300
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.8832799
    alternative = one.sided

NOTE: n is number in *each* group

> (n.1 <- AprxSplSz.prop.test(p, aprx.lv, pwr.at.least, 1))
[1] 320
> power.prop.test(NULL, p[1], p[2], aprx.lv, pwr.at.least, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 319.2656
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.9
    alternative = one.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.1 - 1, n.1 - 1), p, aprx.lv, 1)
[1] 0.8997854
> power.prop.test(n.1 - 1, p[1], p[2], aprx.lv, NULL, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 319
             p1 = 0.8
             p2 = 0.7
```

```
      sig.level = 0.05
          power = 0.8997854
    alternative = one.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.1, n.1), p, aprx.lv, 1)
[1] 0.9005913
> power.prop.test(n.1, p[1], p[2], aprx.lv, NULL, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 320
             p1 = 0.8
             p2 = 0.7
      sig.level = 0.05
          power = 0.9005913
    alternative = one.sided

NOTE: n is number in *each* group

> # alt = "less":
> (p <- c(.7, .8)); (q <- 1 - p); (Delta <- -diff(p))
[1] 0.7 0.8
[1] 0.3 0.2
[1] -0.1
> (pbar <- sum(n*p)/sum(n)); (qbar <- 1 - pbar)
[1] 0.75
[1] 0.25
> (se0 <- sqrt(pbar*qbar*sum(1/n))); (se <- sqrt(sum(p*q/n)))
[1] 0.03535534
[1] 0.03511885
> AprxPwr.prop.test(n, p, aprx.lv, 1)
[1] 0.8832799
> power.prop.test(n0, p[1], p[2], aprx.lv, NULL, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 300
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.8832799
    alternative = one.sided

NOTE: n is number in *each* group

> (n.1 <- AprxSplSz.prop.test(p, aprx.lv, pwr.at.least, 1))
[1] 320
> power.prop.test(NULL, p[1], p[2], aprx.lv, pwr.at.least, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 319.2656
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.9
    alternative = one.sided

NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.1 - 1, n.1 - 1), p, aprx.lv, 1)
[1] 0.8997854
> power.prop.test(n.1 - 1, p[1], p[2], aprx.lv, NULL, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 319
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.8997854
    alternative = one.sided
```

```
NOTE: n is number in *each* group

> AprxPwr.prop.test(c(n.1, n.1), p, aprx.lv, 1)
[1] 0.9005913
> power.prop.test(n.1, p[1], p[2], aprx.lv, NULL, "o") # Aprx

     Two-sample comparison of proportions power calculation

              n = 320
             p1 = 0.7
             p2 = 0.8
      sig.level = 0.05
          power = 0.9005913
    alternative = one.sided

NOTE: n is number in *each* group

> # § 5-6.3 Confidence Interval on the Difference in Binomial Proportions
> # EXAMPLE 5-13 Crankshaft Bearings
> # Refer to EXAMPLES 4-14 and 5-12.
> (n <- c(85, 85)); (x <- c(10, 8)); (Delta0 <- 0); (aprx.lv <- .05)
[1] 85 85
[1] 10  8
[1] 0
[1] 0.05
> (phat <- x/n); (qhat <- 1 - phat); (phat.diff <- -diff(phat))
[1] 0.11764706 0.09411765
[1] 0.8823529 0.9058824
[1] 0.02352941
> (sehat <- sqrt(sum(phat*qhat/n)))
[1] 0.04716252
> (n0 <- mean(n)); (sdhat <- sqrt(n0)*sehat)
[1] 85
[1] 0.4348169
> (ntilde <- n + 2); (xtilde <- x + 1)
[1] 87 87
[1] 11  9
> (ptilde <- xtilde/ntilde); (qtilde <- 1 - ptilde)
[1] 0.1264368 0.1034483
[1] 0.8735632 0.8965517
> (ptilde.diff <- -diff(ptilde))
[1] 0.02298851
> (setilde <- sqrt(sum(ptilde*qtilde/ntilde)))
[1] 0.04832805
> (n0tilde <- mean(ntilde)); (sdtilde <- sqrt(n0tilde)*setilde)
[1] 87
[1] 0.4507741
> # alt = "two.sided":
> prop.test(x, n, NULL, "t", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  x out of n
X-squared = 0.24854, df = 1, p-value = 0.6181
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.06890742  0.11596624
sample estimates:
    prop 1     prop 2
0.11764706 0.09411765

> (CritVal <- qnorm(c(aprx.lv/2, 1 - aprx.lv/2))) # Aprx
[1] -1.959964  1.959964
> (AsymptoticCI <- phat.diff + CritVal*sehat) # Aprx
[1] -0.06890742  0.11596624
> prop.test(xtilde, ntilde, NULL, "t", 1 - aprx.lv, F) # Aprx

        2-sample test for equality of proportions without continuity correction

data:  xtilde out of ntilde
X-squared = 0.22597, df = 1, p-value = 0.6345
alternative hypothesis: two.sided
95 percent confidence interval:
```

```
 -0.07173274  0.11770975
sample estimates:
   prop 1    prop 2
0.1264368 0.1034483

> (AgrestiCaffoCI <- ptilde.diff + CritVal*setilde) # Aprx
[1] -0.07173274  0.11770975
> normal.and.t.dist(NA, NA, phat.diff, sdhat, n0, NA, aprx.lv/2, aprx.lv/2,
+ F, F, T, T, "c") # Aprx
> normal.and.t.dist(NA, NA, ptilde.diff, sdtilde, n0tilde, NA, aprx.lv/2,
+ aprx.lv/2, F, F, T, T, "c") # Aprx
> # EXERCISES FOR § 5-6
> # § 5-7 Summary Tables for Inference Procedures for Two Samples
> # Refer to § 4-9.
> # § 5-8 What if We Have More than Two Samples?
> # § 5-8.1 Completely Randomized Experiment and Analysis of Variance
> # Table 5-5 Tensile Strength of Paper (psi)
> # Figure 5-7 (a) Box plots of hardwood concentration data. (b) Display of
> # the model in equation 5-31 for the completely randomized single-factor
> # experiment.
> # Table 5-6 Typical Data for a Single-Factor Experiment
> # Table 5-7 The Analysis of Variance for a Single-Factor Experiment
> # EXAMPLE 5-14 Tensile Strength
> # Table 5-8 Minitab Analysis of Variance Output for the Paper Tensile
> # Strength Experiment
> # Figure 5-8 Tensile strength averages from the hardwood concentration
> # experiment in relation to a normal distribution with standard deviation
> # sqrt(MS_E/n) = sqrt(6.51/6) = 1.04.
> # Table 5-9 Residuals for the Tensile Strength Experiment
> # Normal Probability Plot Interpretation
> # Figure 5-9 Normal probability plot of residuals from the hardwood
> # concentration experiment.
> # Figure 5-10 Plot of residuals versus factor levels (hardwood
> # concentration).
> # Figure 5-11 Plot of residuals versus \bar{y}_i.
> # § 5-8.2 Randomized Complete Block Experiment
> # Figure 5-12 A randomized complete block design.
> # Table 5-10 A Randomized Complete Block Design
> # Table 5-11 A Randomized Complete Block Design with a Treatments and b
> # Blocks
> # Table 5-12 Analysis of Variance for a Randomized Complete Block Design
> # EXAMPLE 5-15 Fabric Strength
> # Table 5-13 Fabric Strength Data—Randomized Complete Block Design
> # Table 5-14 Analysis of Variance for the Randomized Complete Block
> # Experiment
> # Table 5-15 Minitab Analysis of Variance for the Randomized Complete Block
> # Design in Example 5-15
> # Figure 5-13 Strength averages from the fabric experiment in relation to a
> # normal distribution with standard deviation sqrt(MS_E/b) = sqrt(0.0792/5)
> # = 0.126.
> # Table 5-16 Residuals from the Randomized Complete Block Design
> # Figure 5-14 Normal probability plot of residuals from the randomized
> # complete block design.
> # Figure 5-15 Residuals versus y_{ij}^.
> # Figure 5-16 Residuals by chemical type.
> # Figure 5-17 Residuals by block.
> # EXERCISES FOR § 5-8
>
```